



A unified online test battery for cognitive impulsivity reveals relationships with real-world impulsive behaviours

Antonio Verdejo-García¹✉, Jeggan Tiego¹, Naomi Kakoschke¹, Neda Moskovsky¹, Katharina Voigt¹, Alexandra Anderson¹, Julia Koutoulogenis¹, Dan I. Lubman^{1,2} and Mark A. Bellgrove¹

Impulsive behaviours are a major contributor to the global burden of disease, but existing measures of cognitive impulsivity have suboptimal reliability and validity. Here, we introduce the Cognitive Impulsivity Suite, comprising three computerized/online tasks using a gamified interface. We conceptualize rapid-response impulsive behaviours (disinhibition) as arising from the failure of three distinct cognitive mechanisms: attentional control, information gathering and monitoring/shifting. We demonstrate the construct and criterion validity of the Cognitive Impulsivity Suite in an online community sample ($N = 1,056$), show test-retest reliability and between-subjects variability in a face-to-face community sample ($N = 63$), and replicate the results in a community and clinical sample ($N = 578$). The results support the theoretical architecture of the attentional control, information gathering and monitoring/shifting constructs. The Cognitive Impulsivity Suite demonstrated incremental criterion validity for prediction of real-world, addiction-related problems and is a promising tool for large-scale research on cognitive impulsivity.

Impulsive behaviours are a hallmark of numerous mental disorders, including addictions, personality disorders, bipolar disorder and attention deficit hyperactivity disorder (ADHD)^{1–3}, as well as neurological disorders and related clinical presentations, such as frontotemporal dementia and medication-induced addictive behaviours in Parkinson's disease^{4,5}. Impulsivity is also associated with a range of unhealthy and risk-related behaviours, such as binge drinking, overeating, reckless driving and risky sexual behaviours^{6–8}. Unsurprisingly, impulsivity has been strongly associated with leading causes of morbidity and mortality that contribute to the global burden of disease^{9,10}.

Impulsivity has been classified into separate 'rapid-response' and 'choice' constructs^{11,12}. Rapid-response impulsivity has been defined as the non-reflective selection of stimulus-evoked responses¹¹, and equates to 'disinhibition'¹. Choice impulsivity refers to the tendency to select smaller-sooner rewards over larger-delayed rewards measured with delay discounting tasks¹². There is agreement that the concept of 'disinhibition' can be further fractionated and may arise from the failure of different mechanisms, including attention, reflection and shifting^{13,14}. The function (or dysfunction) of these different mechanisms is at least partially segregated and varies between individuals and contexts⁴. Thus, one or more of these cognitive mechanisms can drive the emergence of impulsive behaviours among different individuals, disorders and situations¹⁵.

Despite evidence that disinhibition is a product of fluctuations in multiple cognitive mechanisms, there are no multidimensional measures of the construct. Cognitive assessment of disinhibition typically uses a single task (for example, the stop signal task) or an assortment of tasks with different origins (for example, some frequently used tasks such as the Stroop were not originally developed to measure impulsivity), format (for example, paper and pencil versus computerized), structure (for example, type of trials or duration)

and basic perceptual and motor demands. These tasks have rarely been subjected to rigorous psychometric assessment, nor were they developed for the purpose of studying individual differences (for example, as a predictor of self-regulation versus impulsive behaviour). As a result, cognitive assessment of disinhibition lacks comparability, reliability and construct validity^{14,16,17}. In contrast, self-report measures, notwithstanding the limitations of this method¹³, are generally developed and optimized as measures of individual differences and undergo rigorous psychometric assessment and construct validation efforts¹⁸. These differences in development and psychometric properties may explain, in part, the lack of convergence between task-based cognitive measures of disinhibition and self-report measures of impulsivity^{18,19}, along with the poor to absent criterion validity of cognitive measures in predicting real-world outcomes, such as mental and physical health problems¹⁶. This lack of convergence has led to calls to re-evaluate the suitability, applicability and scope of task-based assessment of individual differences in cognitive impulsivity and related constructs¹⁹. However, others have suggested that, with careful task design and development, along with attention to psychometric considerations, such as accounting for measurement error, the use of task-based cognitive assessment may provide incremental explanatory power and yield unique mechanistic insights into impulsive behaviours^{20,21}.

Online testing tools and crowdsourcing recruitment and assessment platforms (for example, MTurk) are capable of capturing key performance measures (that is, errors and latency) with high precision and accuracy in diverse and representative samples and therefore provide opportunities for robust psychometric testing of cognitive tasks²². Here, we capitalized on these technologies to develop and test a multidimensional and unified cognitive measure of disinhibition: the Cognitive Impulsivity Suite (CIS). To engineer the CIS, we theorized three distinct cognitive mechanisms

¹School of Psychological Sciences and Turner Institute for Brain and Mental Health, Monash University, Melbourne, Victoria, Australia. ²Turning Point, Eastern Health and Eastern Health Clinical School, Monash University, Melbourne, Victoria, Australia. ✉e-mail: antonio.verdejo@monash.edu

underlying disinhibition: problems of attentional control, lack of information gathering (that is reflection) and reduced feedback monitoring and shifting in response to positive and negative feedback¹⁴. ‘Attentional control’ reflects the ability to energize and sustain cognitive resources to control prepotent responses²³. It has typically been measured with go/no-go, stop signal and the sustained attention to response task (SART)²⁴. ‘Information gathering’ refers to the ability to collect information, and to consider its accuracy, before responding or making a decision¹³. It is usually measured with information sampling and jumping to conclusions (JTC) tasks²⁵. ‘(Feedback) monitoring/shifting’ reflects the ability to monitor the positive and negative outcomes of previous actions and decisions and to adapt ongoing behaviour accordingly¹⁴. It has typically been measured with rule shifting tasks, and it is one of the cognitive mechanisms implicated in the Iowa Gambling Task (IGT)²⁶. We based our proposed model comprising three cognitive processes on meta-analytic factor findings and brain lesion-based architectonic localization studies that have identified separate brain areas and cognitive processes contributing to disinhibition²⁷. We acknowledge the existence of other contemporary models that have proposed multiple cognitive processes based on factor analyses of particular datasets^{16,28,29}, but we sought to combine meta-analytic evidence and biological plausibility in proposing the three selected processes.

Guided by our theoretical conceptualization, we devised three tasks to measure the cognitive mechanisms of interest. We worked with game developers to harmonize perceptual and response demands, such that all three tasks had similar stimuli and types of responses required³⁰. To facilitate participants’ engagement with online testing and reliable measurement during remote assessment sessions, we introduced ‘gaming’ elements (for example, a common theme of the ‘Wild West’, cartoon characters and in-game point schedules). We reasoned that in-game points would also facilitate the capture of motivational/emotional aspects of impulsivity that are factored in by trait measures (for example, sensation seeking, and positive and negative urgency) but not by traditional cognitive tasks, thus contributing to convergent validity. Accuracy/error data obtained from performance of the three tasks were used as the dependent variables to measure the target constructs, rather than reaction time and difference (that is, interference) scores, because the latter have poor reliability, and convergent and criterion validity^{16,17,31–33}. Figure 1 displays the end product: the CIS.

We examined the psychometric properties (that is, reliability and validity) of the CIS across three independently obtained samples. We tested the construct and criterion validity of the CIS in an online community sample recruited through Amazon Mechanical Turk (AMT), which we subsequently split into two samples of equal size (AMT calibration and validation subsamples), and aimed to replicate the results in a community and clinical sample in a pre-registered study/hypotheses (<https://osf.io/vhn95/>). Furthermore, we examined reliability and between-subjects variability in a face-to-face community sample.

We predicted that a three-factor model differentiating between attentional control, information gathering and monitoring/shifting, as largely empirically independent factors, would provide the best fit to performance data obtained on the CIS, supporting the factorial validity and discriminant validity of these three disinhibition constructs. We hypothesized that the attentional control, information gathering and monitoring/shifting factors would be differentially related to the five facets of impulsivity represented by the short Urgency, Lack of Premeditation, Lack of Perseverance, Sensation Seeking, Positive Urgency Scale (sUPPS-P) (ref. ³⁴). It was expected that these empirical relationships, estimated within a structural equation modelling (SEM) framework, would be substantially stronger than those previously reported between cognitive and self-report measures of impulsivity, as indicated by larger

effect size estimates (that is, $r_{es} > 0.1–0.3$) (ref. ³⁵). Additionally, we predicted that the CIS would demonstrate concurrent criterion validity by explaining statistically and practically significant variance in the severity of addiction problems (that is, $R^2 > 0.00$, one-tailed $P < 0.05$). We further predicted that the cognitive impulsivity factors measured by the CIS would demonstrate incremental criterion validity by explaining unique additional variance in addiction problems above that explained by self-reported impulsivity, as indicated by an increase in the proportion of variance explained (that is, a change in the coefficient of variation $\Delta R^2 > 0.00$) once the CIS factors were included as predictors. We also predicted that the CIS would show reliability across repeated testing and modes of administration.

Results

Participants’ feedback. A subset of participants from the community sample ($n = 194$) completed an online survey to collect data on their enjoyment of and engagement with the CIS, as well as their feedback on task duration, difficulty and clarity of instructions. The results are provided in Extended Data Figs. 1–5 and Supplementary Tables 1–5.

Factorial validity. The factorial validity of the CIS was first evaluated in the AMT calibration subsample. We tested the hypothesized three-factor model and then compared this against several theoretically plausible competing models for statistical fit, interpretability and model parsimony. The hypothesized three-factor model of the CIS (Fig. 2) provided a reasonable fit to the data ($\chi^2(47) = 66.238$, $P = 0.067$; root mean square error of approximation (RMSEA) 0.028 (90% CI 0.008–0.043); comparative fit index (CFI) 0.992; standardized root mean residual (SRMR) 0.039). Competing one- and two-factor confirmatory factor analysis (CFA) models failed to provide a better fit to the data than the hypothesized correlated three-factor model (Supplementary Table 8). Model diagnostics were calculated and established discriminant validity between the three factors. These results, along with tests of construct replicability, are provided in Supplementary Information.

The three-factor CIS model was cross-validated in the AMT validation subsample and independent community and clinical sample using invariance testing and the effects coding method for scaling the latent variables³⁶. The results of invariance testing are summarized in Table 1. In summary, the dimensional structure and measurement properties of the three-factor model were largely equivalent across the three samples, indicating that the CIS factors could be meaningfully compared, empirically and conceptually, across the three samples. The one exception was statistical non-invariance of the factor loading matrix for the attentional control factor in the AMT validation subsample. The results indicated that this factor had a different substantive interpretation from the attentional control factor in the other two samples. This can probably be partly attributed to instability in parameter estimation due to a high fraction of missing information (missing data) for no-go commission errors on the bounty hunter task in the AMT validation subsample. Details of construct reliability in the AMT validation subsample are provided in Supplementary Information.

We tested for invariance of the CIS three-factor model between female and male participants from the AMT and community and clinical samples (Supplementary Table 9). In the AMT sample, we found evidence for full configural, weak and strict invariance, with partial strong invariance (freely estimated intercepts for fast response errors on all four blocks of the caravan spotter task) and partial invariance of the factor variance–covariance matrix (variance for the information gathering factor was freely estimated in the male sample). In the community and clinical sample, full invariance of all model parameters was demonstrated except for the residual variance of fast response errors on the caravan spotter task. Results

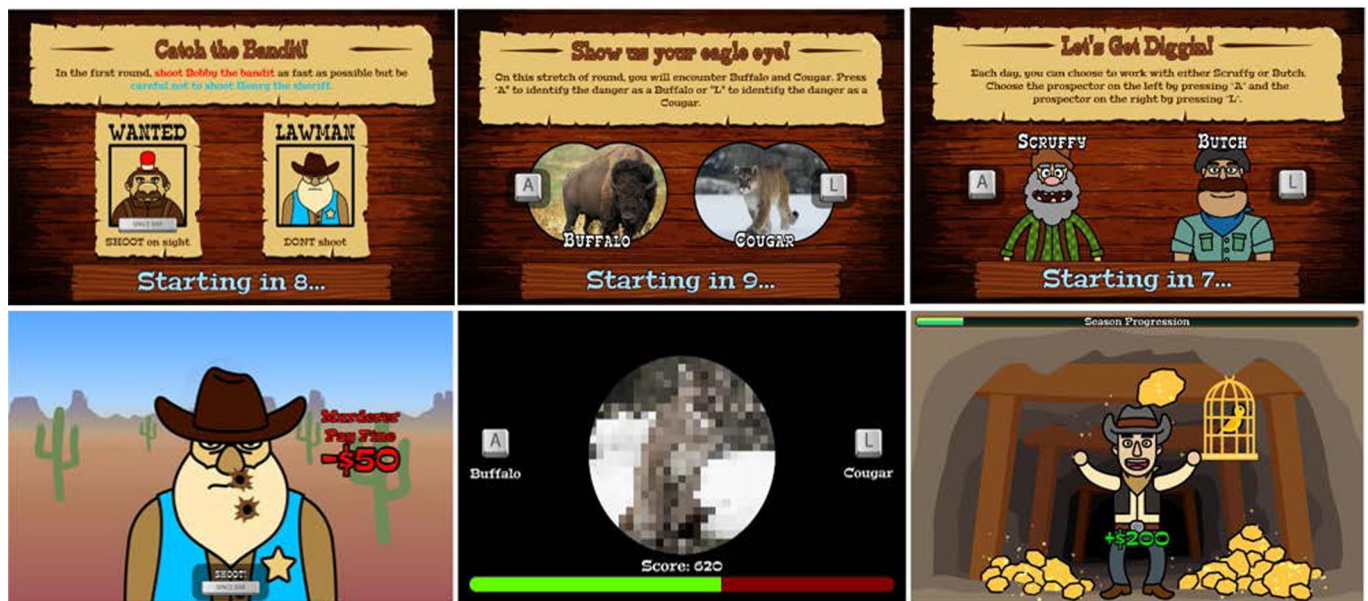


Fig. 1 | The three tasks that form the CIS. CIS tasks description, left to right: bounty hunter (attentional control): participants are presented with one of two stimuli, a bandit or a sheriff; the goal is to shoot the bandits ('go') but not the sheriffs ('no-go'). Sheriffs usually appear in daylight (blue sky, in the figure) and bandits at dusk (dark sky, not shown). Therefore, blue and dark skies are cues that predict higher probability of 'no-go' and 'go' stimuli, respectively. Stimulus onset asynchronies (SOAs, that is, the interval between the cue and the stimulus) are manipulated such that some SOAs are short (50 ms, 200 ms) and some long (1,200 ms, 3,000 ms) to challenge attentional mechanisms. Caravan spotter (information gathering): participants are presented with a totally ambiguous pixelated image that gradually disambiguates over the course of the trial, and instructed to decide which of two stimuli the pixelated image will ultimately morph into (for example, a buffalo or a cougar). Participants are awarded points for speed and accuracy, but penalized for incorrect responses. Prospector's gamble (monitoring/shifting): participants are instructed to choose the luckier of the two prospectors in an (unknown to participants) 75/25 probabilistic reinforcement schedule. The correct/incorrect feedback (that is, indicating whether they have selected/not selected the luckier prospector, that is, the one that secures the reward ~75% of the times), delivered after each trial in basic or enhanced format (see details in Methods; the figure displays an example of rewarded enhanced trial), is designed to promote reward- and loss-driven impulsive responses. Short clips of the three tasks can be seen on YouTube at <https://www.youtube.com/playlist?list=PLs-GMH-Foyaz-UkhauX-uSCDBQkAM1sTv>.

of invariance testing also established the equivalence of the covariance structure for the CIS in the community and clinical sample prior to being combined into one sample as hypothesized in our pre-registration plan (<https://osf.io/vhn95/>). Details are provided in Supplementary Table 10.

Convergent validity: cognitive tasks. The CFA models with the CIS factors and each of the same-family cognitive tasks (SART, JTC and IGT) provided a good fit to the data (Extended Data Figs. 6–8). Adjusted significance thresholds for designating statistical significance based on the adjusted Bonferroni 2 (AB2) procedure³⁷ for each of the cognitive task variables were $\alpha < 0.025$, $\alpha < 0.027$ and $\alpha < 0.024$ for SART, IGT and JTC, respectively. As expected, performance on SART was associated with attentional control ($\phi = 0.287$, s.e. 0.079, 95% bootstrapped confidence interval (BCI) 0.123 to 0.434, $P < 0.001$) and less strongly with monitoring/shifting ($\phi = 0.274$, s.e. 0.068, 95% BCI 0.137 to 0.402, $P < 0.001$) and information gathering ($\phi = 0.176$, s.e. 0.073, 95% BCI 0.027 to 0.312, $P = 0.016$). Similarly, in agreement with our initial assumptions, performance on the IGT was associated with monitoring/shifting ($\phi = 0.222$, s.e. 0.060, 95% BCI 0.107 to 0.338, $P < 0.001$), but not attentional control ($\phi = 0.048$, s.e. 0.077, 95% BCI -0.098 to 0.209, $P = 0.532$) or information gathering ($\phi = 0.070$, s.e. 0.066, 95% BCI -0.057 to 0.206, $P = 0.290$). Performance on the JTC task was associated with attentional control ($\phi = 0.175$, s.e. 0.079, 95% BCI 0.022 to 0.332, $P = 0.027$) and monitoring/shifting ($\phi = 0.152$, s.e. 0.069, 95% BCI 0.017 to 0.286, $P = 0.027$), but these associations did not meet the adjusted threshold for statistical significance ($\alpha < 0.024$). Median number of beads drawn before deciding on the JTC was not meaningfully associated

with the information gathering factor ($\phi = -0.029$, s.e. 0.077, 95% BCI -0.175 to 0.129, $P = 0.708$).

Convergent validity: self-reported impulsivity. The three CIS factors and the five self-reported facets of impulsivity represented by the sUPPS-P scales were combined into one model for all three samples to evaluate the convergent/divergent validity of the attentional control, information gathering and monitoring/shifting factors compared with previous research³⁵. Negative urgency, lack of perseverance, lack of premeditation, sensation seeking and positive urgency were entered into the model as single-indicator latent variables with error variance fixed to reflect the inverse of Cronbach's alpha (α) internal consistency reliability estimate ($\theta_{\epsilon} = [1 - \alpha] \times \sigma$) (ref. ³⁸). The variances and corresponding error variances were calculated separately for each of the three participant groups. The initial fit of the configural model across all three samples was adequate ($\chi^2(313) = 352.054$, $P = 0.211$; RMSEA 0.015, 90% CI 0.000–0.023; CFI 0.995; SRMR 0.033). The unstandardized factor variances and covariances between CIS and sUPPS-P subscale factors could be constrained to equality across groups (except for the variance of lack of perseverance, lack of premeditation and positive urgency, as well as the covariance between attentional control and lack of perseverance, which were freely estimated in the community and clinical sample), without leading to a statistically significant decrement in the overall model fit ($\chi^2(352) = 396.360$, $P = 0.199$; RMSEA 0.015, 90% CI 0.000–0.023; CFI 0.995; SRMR 0.040; $\Delta\chi^2(39) = 44.306$, $P = 0.258$). These results provide strong evidence of the stability and reproducibility of the interrelationships between the CIS factors and sUPPS-P self-reported impulsivity dimensions. The standardized estimates for the factor intercorrelations (ϕ) along

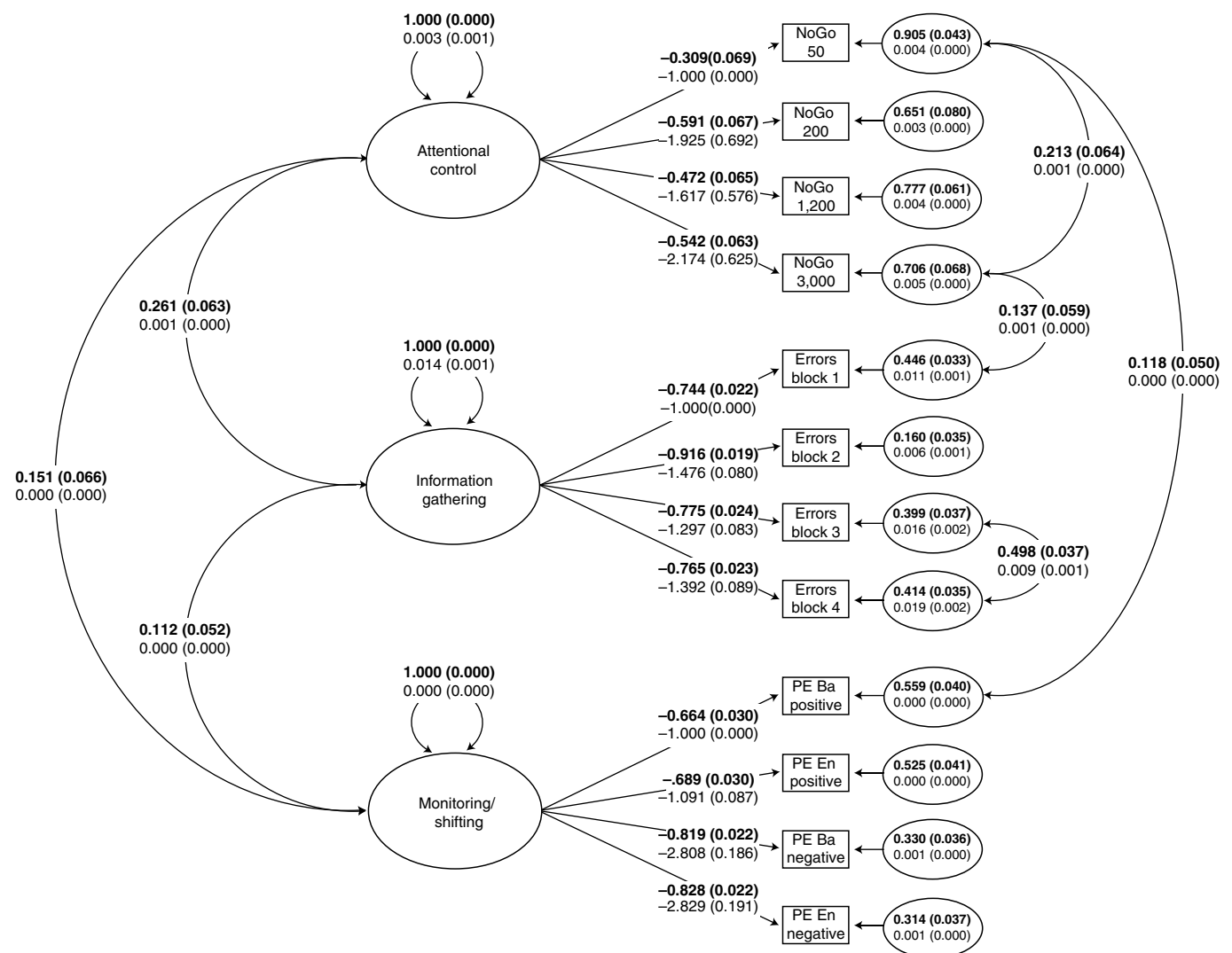


Fig. 2 | Correlated three-factor measurement model of the CIS. Model fit statistics were $\chi^2(47) = 66.238$, $P = 0.067$; RMSEA 0.028 (90% CI 0.008–0.043); CFI 0.991; SRMR 0.039. NoGo 50, commission errors on no-go trials on bounty hunter task with 50 ms stimulus onset asynchrony; NoGo 200, commission errors on no-go trials on bounty hunter task with 200 ms stimulus onset asynchrony; NoGo 1,200, commission errors on no-go trials on bounty hunter task with 1,200 ms stimulus onset asynchrony; NoGo 3,000, commission errors on no-go trials on bounty hunter task with 3,000 ms stimulus onset asynchrony; Errors block 1, fast identification errors on caravan spotter task block 1; Errors block 2, fast identification errors on caravan spotter task block 2; Errors block 3, fast identification errors on caravan spotter task block 3; Errors block 4, fast identification errors on caravan spotter task block 4; PE Ba positive, perseverative errors on prospector's gamble task with basic positive feedback on previous trial; PE En positive, perseverative errors on prospector's gamble task with enhanced positive feedback on previous trial; PE Ba negative, perseverative errors on prospector's gamble task with basic negative feedback on previous trial; PE En negative, perseverative errors on prospector's gamble task with enhanced negative feedback on previous trial. Factor scaling was performed using the reference variable method. Standardized estimates appear in bold typeface. Unstandardized estimates are in normal typeface below. Bootstrapped standard errors (10,000 posterior draws) appear in brackets. Probability values for standardized estimates of the four error covariances were initially estimated in the unconstrained three-factor model and adjusted (B-H $P = 0.02$) for multiple post hoc comparisons using the false discovery rate ($q = 0.05$). $n = 517$.

with the 95% BCIs for each of the three samples are presented in Table 2. The significance thresholds for the factor intercorrelations were adjusted for multiple comparisons using the AB2 procedure, taking into account average parameter dependencies within the model for each sample (Table 2)³⁷.

The hypothesis that the information gathering factor would exhibit stronger negative correlations with sensation seeking and positive urgency than the other three self-reported factors of impulsivity represented in the sUPPS-P was supported. The strength of the correlations between information gathering with sensation seeking ranged between $\phi = -0.197$ and -0.234 , and between information gathering and positive urgency factor estimates were between

$\phi = -0.197$ and -0.275 . Correlations with the other sUPPS-P impulsivity factors ranged between $\phi = -0.068$ and -0.113 . The hypothesis of stronger relationships between monitoring/shifting with positive urgency, negative urgency and lack of premeditation was also supported, in that observed latent correlations between these factors were $\phi = -0.081$ to -0.280 , whereas the linear associations with lack of perseverance and sensation seeking were all weak ($\phi = -0.008$ to -0.046) and did not meet the adjusted threshold for statistical significance. Finally, the predicted stronger relationship between attentional control with positive urgency, compared with the other sUPPS-P impulsivity factors within each sample, was also largely supported, with the exception of the stronger association

Table 1 | Results of invariance testing for CIS three-factor model and addiction problems using the effects coding method across the AMT calibration and validation subsamples and the community and clinical sample

Model	df	χ^2	P	RMSEA (90% CI)	CFI	SRMR	Δ df	$\Delta\chi^2$	ΔP
CIS ^a									
Configural invariance	145	166.177	0.217	0.017 (0.000–0.027)	0.997	0.031			
Partial weak invariance ^b	159	189.699	0.138	0.019 (0.002–0.029)	0.995	0.034	14	23.522	0.052
Partial strong invariance ^c	174	213.999	0.078	0.021 (0.009–0.030)	0.994	0.036	15	24.300	0.060
Partial strict invariance ^d	189	233.753	0.072	0.021 (0.010–0.030)	0.993	0.037	15	19.754	0.182
Partial invariance of factor variances and covariances ^e	195	241.310	0.070	0.021 (0.010–0.030)	0.993	0.042	6	7.557	0.272
Addiction problems ^f									
Configural invariance	0	0.000	0.000	0.000 (0.000–0.000)	0.000	0.000			
Weak invariance	4	2.664	0.723	0.000 (0.000–0.055)	1.000	0.013	4	2.664	0.616
Partial strong invariance ^g	6	5.217	0.619	0.000 (0.000–0.053)	1.000	0.017	2	2.553	0.279
Partial strict invariance ^h	11	10.046	0.801	0.000 (0.000–0.043)	1.000	0.030	5	4.829	0.437
Invariance of factor variance	13	10.074	0.864	0.000 (0.000–0.034)	1.000	0.030	2	0.028	0.986

Notes: df, degrees of freedom; χ^2 , chi-squared value for test of model fit using maximum-likelihood estimation with expectation maximization; P, significance value of the chi-squared test statistic with Bollen–Stine bootstrapping (10,000 posterior draws); Δ df, delta degrees of freedom; $\Delta\chi^2$, delta chi squared; ΔP , significance value of the delta chi-squared test statistic. ^aAMT calibration subsample, $n=517$; AMT validation subsample, $n=499$; community and clinical sample, $n=569$. ^bUnstandardized loadings on the attentional control factor freely estimated for no-go commission errors on all four SOAs from the bounty hunter task in the AMT calibration subsample and no-go commission errors at 50 ms SOA in the community and clinical sample. ^cIntercepts freely estimated for no-go commission errors at 200 ms SOA from the bounty hunter task in the AMT validation subsample, and for no-go commission errors at 50 ms SOA in the community and clinical sample, along with fast response errors on block 1 of the caravan spotter task. ^dUnstandardized residual variances of no-commission errors at SOA 50, 200 and 1,200 ms freely estimated in the AMT validation subsample and community and clinical sample, as well as (in the latter) the residual variances and error covariance for fast response errors blocks 2 and 4, and all of the prospector's gamble task variables except for perseverative errors with basic positive feedback. ^eUnstandardized variance for attentional control, as well as its covariance with information gathering and monitoring/switching, freely estimated in the AMT validation subsample, and unstandardized variance of the attentional control and information gathering factors, as well as their covariance, freely estimated in the community and clinical sample. ^fAMT calibration subsample, $n=489$; AMT validation subsample, $n=497$; community and clinical sample, $n=577$. The unstandardized variance estimates for the addiction problems factor in the configural model were: AMT calibration subsample $\phi = 0.255$, s.e. 0.039, 95% bootstrapped confidence interval (BCI) 0.211–0.308, $P < 0.001$; AMT calibration subsample $\phi = 0.269$, s.e. 0.027, 95% BCI 0.219–0.327, $P < 0.001$; community and clinical sample $\phi = 0.255$, s.e. 0.024, 95% BCI 0.207–0.303, $P < 0.001$. ^gAll three intercepts freely estimated in the community and clinical sample. ^hThe unstandardized residual variance for AUDIT was freely estimated in the community and clinical sample. Latent mean differences were calculated for the information gathering ($Z = 0.866$, $P = 0.386$) and monitoring/switching ($Z = 0.358$, $P = 0.720$) factor in the AMT validation subsample, and for the monitoring/switching ($Z = 1.598$, $P = 0.110$) factor in the community and clinical sample. Latent mean differences were calculated for the addiction problems factor between the AMT calibration and validation subsamples ($Z = -0.376$, $P = 0.707$).

with lack of premeditation ($\phi = -0.215$) than with positive urgency ($\phi = -0.176$) in the community and clinical sample.

Criterion validity. A one-factor model combining alcohol, drug and gambling problems, with age and years of education included in the model as covariates, provided the most theoretically and empirically parsimonious representation of addiction problems in the AMT calibration subsample ($\chi^2(4) = 12.630$, $P = 0.098$; RMSEA 0.066, 90% CI 0.027–0.190; CFI 0.946; SRMR 0.029) (Supplementary Table 12). This one-factor model was replicated in the AMT validation subsample and community and clinical sample with invariance testing (Table 1). Further details of the evaluation of the initial model specification and the evaluation of the model fit can be found in Supplementary Information. The threshold for determining statistical significance of associations between the addiction problems and CIS factors was adjusted within each sample using the AB2 procedure (AMT calibration, $\alpha = 0.25$; AMT validation, $\alpha = 0.027$; community and clinical, $\alpha = 0.029$). The addiction problems factor was modestly associated with monitoring/shifting in the AMT subsamples (AMT calibration, $\phi = -0.243$, s.e. 0.056, 95% BCI -0.358 to -0.138 , $P < 0.001$; AMT validation, $\phi = -0.234$, s.e. 0.066, 95% BCI -0.366 to -0.108 , $P < 0.001$), but not the community and

clinical sample ($\phi = -0.098$, s.e. 0.059, 95% BCI -0.214 to 0.020 , $P = 0.100$). Conversely, the addiction problems factor was moderately strongly associated with information gathering in the community and clinical sample ($\phi = -0.338$, s.e. 0.054, 95% BCI -0.454 to -0.241 , $P < 0.001$), but not the AMT subsamples (AMT calibration, $\phi = 0.005$, s.e. 0.066, 95% BCI -0.126 to 0.132 , $P = 0.942$; AMT validation, $\phi = -0.051$, s.e. 0.069, 95% BCI -0.191 to 0.079 , $P = 0.460$). Finally, attentional control exhibited a weak linear association with addiction problems in the AMT validation subsample ($\phi = -0.164$, s.e. 0.034, 95% BCI -0.337 to -0.032 , $P = 0.034$) and a modest strength association in the community and clinical sample ($\phi = -0.263$, s.e. 0.066, 95% BCI -0.406 to -0.148 , $P < 0.001$), but not in the AMT calibration subsample ($\phi = -0.046$, s.e. 0.088, 95% BCI -0.222 to 0.124 , $P = 0.603$).

A structural regression model was specified in which the addiction problems factor was regressed onto the monitoring/shifting factor in the AMT calibration ($\gamma = -0.243$, s.e. 0.056, 95% BCI -0.358 to -0.138 , $P < 0.001$) and validation ($\gamma = -0.238$, s.e. 0.066, 95% BCI -0.371 to -0.112 , $P < 0.001$) subsamples, and onto the information gathering ($\gamma = -0.277$, s.e. 0.062, 95% BCI -0.402 to -0.162 , $P < 0.001$) and attentional control ($\gamma = -0.160$, s.e. 0.074, 95% BCI -0.309 to -0.025 , $P = 0.029$) factors in the community

Table 2 | Factor intercorrelations between the CIS and sUPPS-P

Variables	Negative urgency	Lack of perseverance	Lack of premeditation	Sensation seeking	Positive urgency
AMT calibration subsample					
Attentional control	−0.115' (−0.200 to −0.073)	−0.060 (−0.134 to 0.007)	−0.071 (−0.141 to −0.009)	−0.078 (−0.160 to −0.028)	−0.212' (−0.332 to −0.181)
Information sampling	−0.096' (−0.151 to −0.038)	−0.078 (−0.153 to −0.011)	−0.113' (−0.180 to −0.045)	−0.197' (−0.249 to −0.133)	−0.197' (−0.259 to −0.147)
Monitoring/shifting	−0.109' (−0.171 to −0.049)	−0.010 (−0.081 to 0.059)	−0.119' (−0.196 to −0.061)	−0.046 (−0.109 to 0.014)	−0.238' (−0.302 to −0.189)
AMT validation subsample					
Attentional control	−0.219' (−0.316 to −0.124)	−0.114 (−0.217 to 0.011)	−0.136 (−0.230 to −0.015)	−0.149' (−0.260 to −0.045)	−0.404' (−0.517 to −0.320)
Information sampling	−0.096' (−0.151 to −0.038)	−0.078 (−0.153 to −0.011)	−0.113' (−0.180 to −0.045)	−0.197' (−0.249 to −0.133)	−0.197' (−0.259 to −0.147)
Monitoring/shifting	−0.109' (−0.171 to −0.049)	−0.010 (−0.081 to 0.059)	−0.119' (−0.196 to −0.061)	−0.046 (−0.109 to 0.014)	−0.238' (−0.302 to −0.189)
Community and clinical sample					
Attentional control	−0.081' (−0.141 to −0.052)	−0.031 (−0.071 to 0.004)	−0.215' (−0.321 to −0.124)	−0.055 (−0.115 to −0.020)	−0.176' (−0.281 to −0.149)
Information sampling	−0.114' (−0.178 to −0.046)	−0.068 (−0.135 to −0.010)	−0.092' (−0.148 to −0.037)	−0.234' (−0.299 to −0.159)	−0.275' (−0.362 to −0.207)
Monitoring/shifting	−0.109' (−0.171 to −0.049)	−0.008 (−0.060 to 0.044)	−0.081' (−0.135 to −0.042)	−0.046 (−0.109 to 0.014)	−0.280' (−0.355 to −0.222)

Notes: 95% BCI in parenthesis (10,000 posterior draws). Adjusted significance thresholds using the AB2 procedure were as follows: AMT calibration subsample, $\alpha < 0.013$; AMT validation subsample, $\alpha < 0.014$; community and clinical sample, $\alpha < 0.011$. * $p < \alpha_{\text{per test}} = \frac{\alpha_{\text{familywise}}}{k - \sqrt{|T|}}$

and clinical sample. (Note that the residual association of the addiction problems factor with attentional control ($\psi = -0.087$, s.e. 0.076, $P = 0.253$) was very weak after regression onto the monitoring/shifting factor ($\gamma = -0.234$, s.e. 0.063, $P < 0.001$) in the AMT validation subsample, and attempts to regress the addiction problems factor onto attentional control as well resulted in model non-convergence.) This model provided an adequate fit to the data ($\chi^2(289) = 352.390$, $P = 0.058$; RMSEA 0.020, 90% CI 0.011–0.027; CFI 0.991; SRMR 0.039) that was far more parsimonious than the saturated measurement model in which all parameters between the addiction problems and CIS factors were estimated ($\Delta\chi^2(5) = 1.814$, $P = 0.874$).

Equality constraints from the previous invariance models for the factor variance–covariance matrix were applied to the CIS and addiction problems factors. Additionally, the regression coefficient between the monitoring/shifting and addiction problems factors was constrained to be equal across the AMT subsamples. The constrained structural regression model provided an adequate and more parsimonious fit to the data ($\chi^2(298) = 359.834$, $P = 0.064$; RMSEA 0.020, 90% CI 0.011–0.027; CFI 0.991; SRMR 0.041; $\Delta\chi^2(9) = 7.444$, $P = 0.591$). The regression of the addiction problems factor on the monitoring/shifting factor explained 5.8% of the variance in the AMT calibration and validation subsamples ($R^2 = 0.058$, s.e. 0.022, one-tailed $P = 0.004$). The regression of the addiction problems factor on the information gathering and attentional control factors explained 12.1% of the variance in the community and clinical sample ($R^2 = 0.121$, s.e. 0.037, one-tailed $P < 0.001$). The structural regression models in the AMT calibration subsample and community and clinical sample are shown in Figs. 3 and 4, respectively.

Incremental criterion validity: AMT subsamples. Details of the initial model specification and the evaluation of fit for the incremental validity models, along with the factor intercorrelations (Supplementary Table 13), can be found in Supplementary Information. The addiction problems factor was regressed onto negative urgency, lack of perseverance, lack of premeditation,

sensation seeking and positive urgency, and the residual covariances with the monitoring/shifting factor freely estimated in both AMT subsamples. The regression coefficients for the addiction problems factor on lack of perseverance ($\gamma = -0.155$, s.e. 0.162, 95% BCI −0.279 to 0.202, $P = 0.338$) in the AMT calibration subsample, and lack of perseverance ($\gamma = 0.061$, s.e. 0.117, 95% BCI −0.279 to 0.202, $P = 0.601$), lack of premeditation ($\gamma = -0.082$, s.e. 0.136, 95% BCI −0.677 to 0.450, $P = 0.545$) and positive urgency ($\gamma = -0.058$, s.e. 0.271, 95% BCI −0.677 to 0.450, $P = 0.829$) were weak in the AMT validation subsample. These weak parameters could be constrained to zero and the regression coefficients for the addiction problems factor on negative urgency and sensation seeking factors constrained to equality across subsamples without leading to a significant decrement in model fit ($\Delta\chi^2(6) = 3.661$, $P = 0.722$). These models explained 25.3% and 33.8% of the variance in the addiction problems factor in the AMT calibration ($R^2 = 0.253$, s.e. 0.046, one-tailed $P < 0.001$) and validation ($R^2 = 0.338$, s.e. 0.053, one-tailed $P < 0.001$) subsamples, respectively.

Regressing the addiction problems factor onto the monitoring/shifting factor and constraining this parameter to equality across groups provided a well-fitting and more parsimonious model ($\chi^2(107) = 125.537$, $P = 0.330$; RMSEA 0.018, 90% CI 0.000–0.030; CFI 0.995; SRMR 0.033; $\Delta\chi^2(1) = 2.822$, $P = 0.093$). The monitoring/shifting factor explained an additional 1.3% of the variance in the addiction problems factor in the AMT calibration subsample ($R^2 = 0.266$, s.e. 0.046, one-tailed $P < 0.001$) and 2.0% additional variance in the AMT validation subsample ($R^2 = 0.358$, s.e. 0.052, one-tailed $P < 0.001$). The final model for the AMT calibration subsample is displayed in Extended Data Fig. 9.

Incremental criterion validity: community and clinical sample. Incremental criterion-related validity of the CIS was tested independently in the community and clinical sample. The initial CFA model incorporating the information gathering, monitoring/shifting, addiction problems and five sUPPS-P subscale factors provided

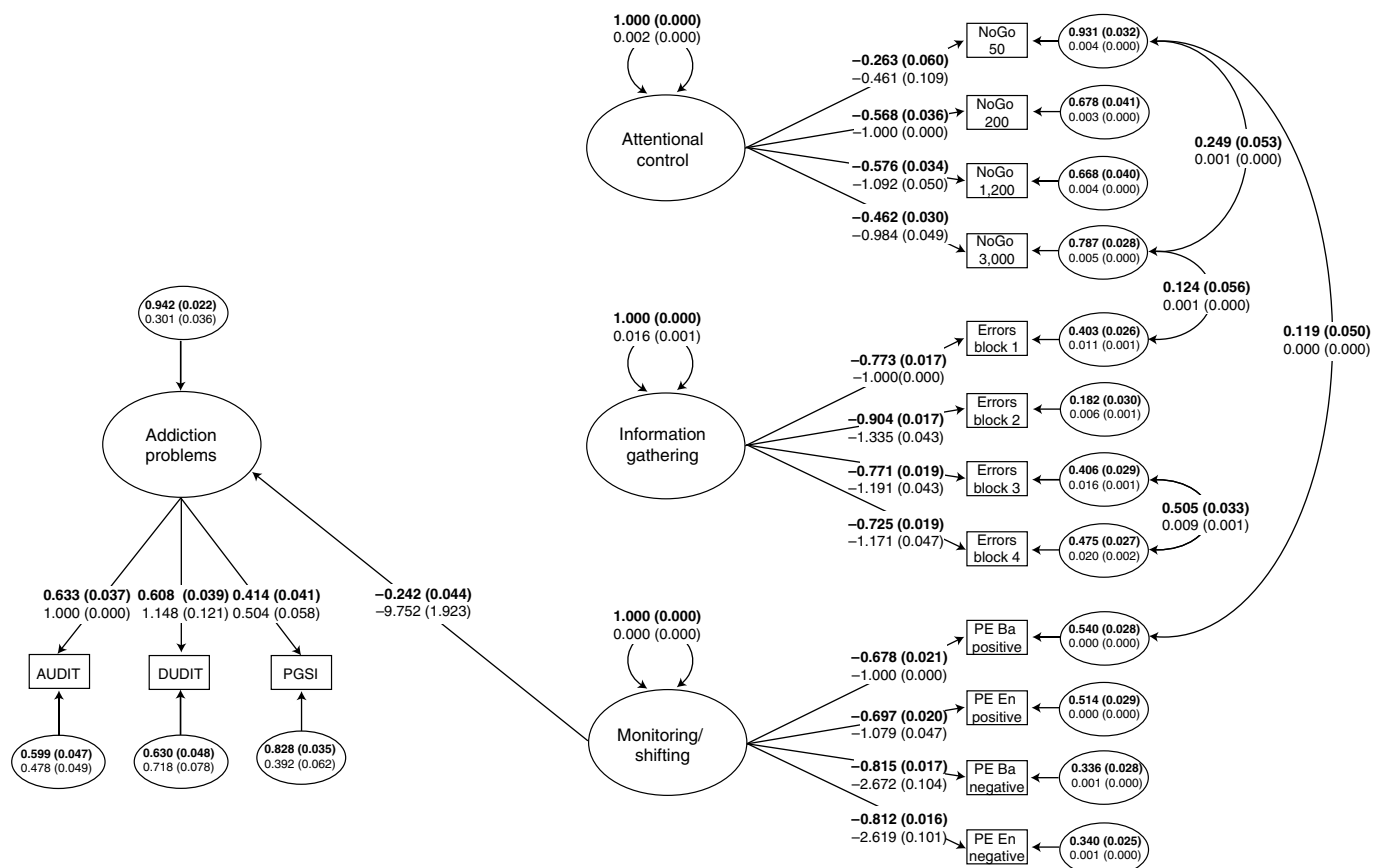


Fig. 3 | Addition problems factor regressed onto the monitoring/shifting factor in the AMT calibration subsample. $\chi^2(298) = 359.834$, $P = 0.064$; RMSEA 0.020, 90% CI 0.011–0.027; CFI 0.991; SRMR 0.041. $R^2 = 0.058$, s.e. 0.022, one-tailed $P = 0.004$. $n = 521$. NoGo 50, commission errors on no-go trials on bounty hunter task with 50 ms stimulus onset asynchrony (SOA); NoGo 200, commission errors on no-go trials on bounty hunter task with 200 ms SOA; NoGo 1,200, commission errors on no-go trials on bounty hunter task with 1,200 ms SOA; NoGo 3,000, commission errors on no-go trials on bounty hunter task with 3,000 ms SOA; Errors block 1, fast identification errors on caravan spotter task block 1; Errors block 2, fast identification errors on caravan spotter task block 2; Errors block 3, fast identification errors on caravan spotter task block 3; Errors block 4, fast identification errors on caravan spotter task block 4; PE Ba positive, perseverative errors on prospector's gamble task with basic positive feedback on previous trial; PE En positive, perseverative errors on prospector's gamble task with enhanced positive feedback on previous trial; PE Ba negative, perseverative errors on prospector's gamble task with basic negative feedback on previous trial; PE En negative, perseverative errors on prospector's gamble task with enhanced negative feedback on previous trial. AUDIT, Alcohol Use Disorder Identification Test raw scores logarithmic transformed; DUDIT, Drug Use Disorder Identification Test raw scores logarithmic transformed; PGSI, Problem Gambling Severity Index raw scores logarithmic transformed. Standardized estimates appear in bold typeface, with unstandardized estimates in normal typeface below. Bootstrapped standard errors (10,000 posterior draws) appear in brackets. The four freely estimated error covariances statistically significant B-H $P = 0.028$. CIS factor intercorrelations omitted for clarity: attentional control with information gathering ($\phi = 0.337$, s.e. 0.045, 95% BCI 0.264–0.440, $P < 0.001$) and monitoring/shifting ($\phi = 0.173$, s.e. 0.064, 95% BCI 0.051–0.300, $P = 0.007$); information gathering with monitoring/shifting ($\phi = 0.099$, s.e. 0.027, 95% BCI 0.048–0.156, $P < 0.001$).

a good fit to the data ($\chi^2(78) = 93.562$, $P = 0.175$; RMSEA 0.019, 90% CI 0.000–0.031; CFI 0.992; SRMR 0.028). The corrected threshold for determining statistical significance of the structural model parameters (that is, factor covariances and regression coefficients) was AB2 $\alpha < 0.020$. Factor intercorrelations are summarized in Supplementary Table 13. The addition problems factor was modestly correlated negatively with attentional control and positively with lack of perseverance, and negatively moderately strongly correlated with information gathering and positively with the remaining self-reported impulsivity factors.

As a first step, the addition problems factor was regressed onto the five self-reported impulsivity factors. The regression coefficients of addition problems on lack of perseverance ($\gamma = 0.092$, s.e. 0.075, 95% BCI -0.055 to 0.240 , $P = 0.224$) and positive urgency ($\gamma = 0.089$, s.e. 0.135, 95% BCI -0.184 to 0.348 , $P = 0.508$) were weak and could be constrained to zero without harming the model fit ($\Delta\chi^2(2) = 1.904$, $P = 0.386$). This model explained

43.1% of the variance in the addition problems factor ($R^2 = 0.431$, s.e. 0.057, one-tailed $P < 0.001$). The residual correlation with the information gathering factor was modest ($\phi = -0.215$, s.e. 0.067, 95% BCI -0.356 to -0.091 , $P = 0.001$) but very weak with attentional control ($\phi = -0.104$, s.e. 0.067, 95% BCI 0.046 to -0.276 , $P = 0.201$), and this latter parameter could be constrained to zero without a significant decrement in the model fit ($\Delta\chi^2(1) = 1.813$, $P = 0.178$). Given the lack of contribution to explaining variance in addition problems, the attentional control factor was removed and the model re-estimated. Model parameters were highly stable with removal of the attentional control factor, suggesting an absence of interpretational confounding³⁹. The model with attentional control removed and the addition problems factor regressed on negative urgency, lack of premeditation and sensation seeking provided a good fit to the data ($\chi^2(39) = 57.631$, $P = 0.081$; RMSEA 0.029, 90% CI 0.010–0.044; CFI 0.988; SRMR 0.028) and explained a moderate proportion of the variance in the addition problems

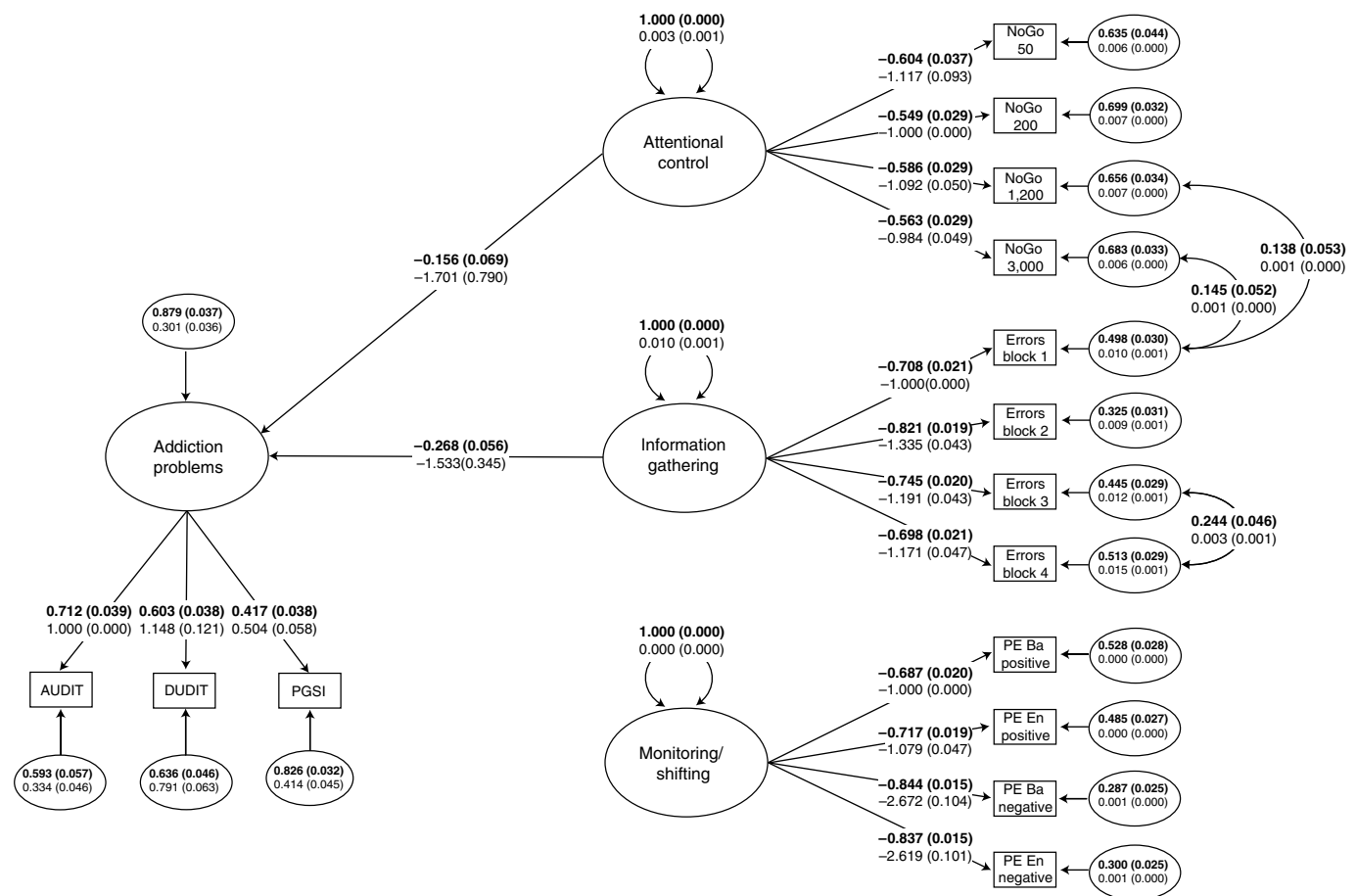


Fig. 4 | Addiction problems factor regressed onto the information gathering and attentional control factors in the community and clinical sample.

$\chi^2(298) = 359.834$, $P = 0.064$; RMSEA 0.020, 90% CI = 0.011–0.027; CFI 0.991; SRMR 0.041. $R^2 = 0.121$, s.e. 0.037, one-tailed $P < 0.001$. $n = 578$. NoGo 50, commission errors on no-go trials on bounty hunter task with 50 ms stimulus onset asynchrony (SOA); NoGo 200, commission errors on no-go trials on bounty hunter task with 200 ms SOA; NoGo 1,200, commission errors on no-go trials on bounty hunter task with 1,200 ms SOA; NoGo 3,000, commission errors on no-go trials on bounty hunter task with 3,000 ms SOA; Errors block 1, fast identification errors on caravan spotter task block 1; Errors block 2, fast identification errors on caravan spotter task block 2; Errors block 3, fast identification errors on caravan spotter task block 3; Errors block 4, fast identification errors on caravan spotter task block 4; PE Ba positive, perseverative errors on prospector's gamble task with basic positive feedback on previous trial; PE En positive, perseverative errors on prospector's gamble task with enhanced positive feedback on previous trial; PE Ba negative, perseverative errors on prospector's gamble task with basic negative feedback on previous trial; PE En negative, perseverative errors on prospector's gamble task with enhanced negative feedback on previous trial; AUDIT, Alcohol Use Disorder Identification Test raw scores logarithmic transformed; DUDIT, Drug Use Disorder Identification Test raw scores logarithmic transformed; PGSI, Problem Gambling Severity Index raw scores logarithmic transformed. Standardized estimates appear in bold typeface, with unstandardized estimates in normal typeface below. Bootstrapped standard errors (10,000 posterior draws) appear in brackets. The three freely estimated error covariances were statistically significant B–H $P = 0.012$. CIS factor intercorrelations omitted for clarity: attentional control with information gathering ($\phi = 0.304$, s.e. 0.040, 95% BCI 0.237–0.392, $P < 0.001$) and monitoring/shifting ($\phi = 0.366$, s.e. 0.050, 95% BCI 0.277–0.473, $P < 0.001$); information gathering with monitoring/shifting ($\phi = 0.123$, s.e. 0.035, 95% BCI 0.060–0.194, $P < 0.001$).

factor ($R^2 = 0.432$, s.e. 0.058, one-tailed $P < 0.001$). Regression of the addiction problems factor onto the information gathering factor explained an additional 2.8% of the variance to that explained by self-reported impulsivity ($R^2 = 0.460$, s.e. 0.057, one-tailed $P < 0.001$; $\Delta R^2 = 0.028$). This final model is displayed in Extended Data Fig. 10. We also conducted sensitivity analyses incorporating age and education into the models, as well as running multiple imputation to test for parameter bias due to missing data not at random. In brief, none of the main findings were substantively different. Details can be found in Supplementary Information.

Face-to-face test–retest reliability. Figure 5 shows the relative proportion of each variance component for each variable of interest. The dominant component of variance for all variables was between-subjects variance, that is, systematic differences in performance between participants, which suggests that the CIS is sensitive

to analysis of individual differences, unlike most previous cognitive impulsivity tasks as evinced by recent work¹⁷. The intraclass correlation coefficient (ICC) estimates for all (but one) variables of interest exceeded the threshold indicating good/substantial reliability (> 0.60) (ref. 39). The accuracy variables of two tasks (bounty hunter and caravan spotter) both showed particularly strong reliability of 0.71. The fast identification errors measure from the caravan spotter task was the only variable that showed relatively modest reliability (0.52). Further, all variables had either no or minimal within-subjects variance, indicating that participants performed consistently between the two administrations. The 95% confidence intervals for the ICCs are reported in Supplementary Table 17.

Equivalence between face-to-face and online administrations. Performance on each of the CIS tasks was similar across the face-to-face and online AMT samples (Table 3). Although the proportion

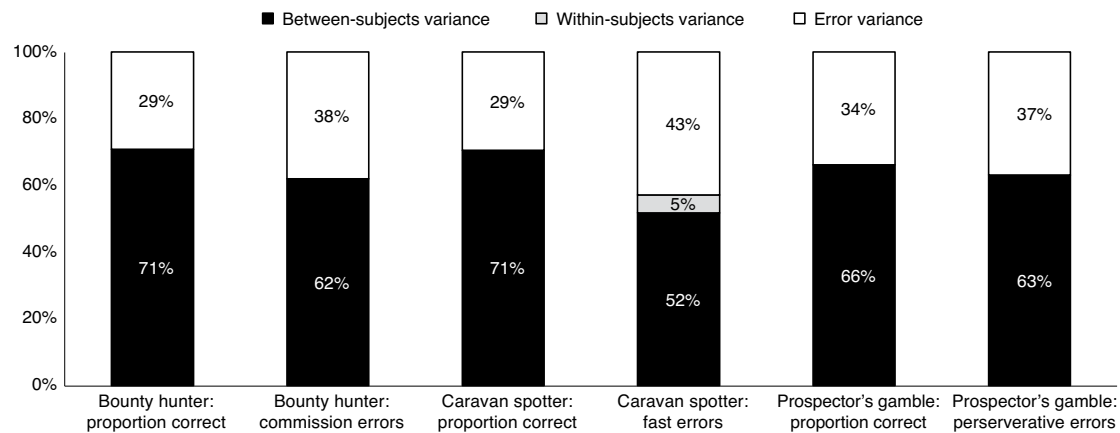


Fig. 5 | Total variance separated into between-subjects, within-subjects and error variance, for performance variables of the CIS. The size of the bars indicates the relative proportion of variance components compared with total variance. For all variables except caravan spotter: fast errors, within-subjects variance estimates were negative, which we replaced with zeros and so are not shown in this figure.

Table 3 | Means and standard deviations for CIS variables for the face-to-face and online AMT samples

CIS variable	Face-to-face sample	Online sample	<i>t</i>	<i>df</i>	<i>P</i>
	<i>M</i> (s.d.)	<i>M</i> (s.d.)			
Attentional control					
Proportion correct	0.975 (0.032)	0.973 (0.032)	0.498	1,047	0.618
Commission errors	0.018 (0.023)	0.014 (0.016)	10.28	67.11	0.206
Information gathering					
Proportion correct	0.881 (0.101)	0.856 (0.110)	2.454	79.40	0.016
Fast errors	0.052 (0.065)	0.062 (0.092)	−1.28	82.22	0.205
Monitoring/shifting					
Proportion correct	0.711 (0.110)	0.688 (0.115)	1.69	1,059	0.106
Perseverative errors	0.281 (0.106)	0.303 (0.110)	−1.62	1,057	0.105

Note: *M*, mean; bold denotes significant *P* value.

of correct responses for the caravan spotter task was higher in the face-to-face than online sample, such a difference would not survive correction for multiple comparisons, that is, Benjamini–Hochberg (B–H) $q=0.05$, $P=0.618$.

Discussion

We provide evidence for the hypothesized factor structure, reliability and validity of the CIS, a unified test battery for disinhibition, which is a major contributor to global disease burden^{9,10}. The performance indices of the three CIS tasks showed consistency across samples, repeated administrations and modes of delivery, including face-to-face and online testing. The CIS indices exhibited a pattern of empirical associations with self-reported impulsivity that was theoretically consistent and larger in magnitude than those previously reported^{16,35}. Performance on the three CIS tasks also generally converged with cognitive measures of the same thematic network, and with real-world outcomes, namely addiction problems. However, the relationships between CIS indices and addiction problems deviated from our pre-registered hypothesis, as we expected associations with monitoring/shifting (not information gathering). Importantly though, the CIS explained additional unique variance in addiction problems to that explained by self-reported impulsivity. Although the percentage of additional variance explained was seemingly small, this can be interpreted in the context of common method variance between self-reported impulsivity and addiction

problems⁴⁰, and the stronger relationships between the cognitive impulsivity factors and self-reported impulsivity than previously reported³⁵, resulting in less unique variance to be explained by the CIS. The obtained effect sizes are also relatively large when interpreted in the context of the total explained variance generally observed in the social and behavioural sciences^{41,42} and, moreover, the minimal amounts of variance usually explained by cognition in self-reported criterion measures^{16,19–21,35}.

CFA demonstrated the strong psychometric properties of the CIS, including a clear three-factor structure and measurement invariance of most model parameters across three independent samples, suggesting psychometric stability of the three tasks and replicability of the findings. The three CIS factors exhibited modest linear associations with other cognitive tasks traditionally used in impulsivity research. The pattern of correlations was mostly consistent with original assumptions, as attentional control and monitoring/shifting were most strongly correlated with SART and IGT, respectively. Information gathering and monitoring/shifting were also correlated with the SART, although less strongly. These associations between the CIS factors and performance on these traditional tasks explained between 3.1% and 8.2% of shared variance. These estimates are similar in magnitude to those generally reported in studies of the convergence between complex cognitive processes and performance on multicomponential tasks⁴³. Unexpectedly, information gathering did not correlate with performance on the

JTC task, a traditional measure of information sampling. Lack of correlations may be due to different demands of tasks. For example, the caravan spotter task requests participants to respond within a short time frame, whereas the JTC task allows participants to sample information without time limits¹¹. This novel feature of the CIS is important as the notion of rapid, unplanned reactions is inherent to the definition of impulsivity¹¹. It was also unexpected that both information gathering and monitoring/shifting correlated with the SART. One potential explanation is that existing response inhibition tasks such as the SART or stop signal are too complex to be ascribed to one particular aspect of impulsivity¹⁴.

The strength of the correlations between the three CIS factors and self-reported impulsivity exceeded previous reports. A meta-analysis of the association between the five facets of self-reported impulsivity and task-based measures of impulsivity reported a mean sample-size-weighted effect size of $r=0.100$, reflecting a mere 1% empirical overlap³⁵. Correlations between task performance and specific facets of impulsivity did not exceed $r=0.13$. By contrast, correlations between the attentional control, information gathering and monitoring/shifting constructs of the CIS and self-reported impulsivity had effect sizes (r) generally on the order of 0.2–0.3, reflecting greater than approximately 4–5% shared variance. The previously reported weak associations may be attributable to low reliability of cognitive measures of impulsivity¹⁷. The homogeneous perceptual and response demands, in-game affective feedback and accuracy/error metrics are core features of the CIS that we anticipated could improve associations with trait measures^{14,16,31,33}. Higher observed estimates in the current study are at least partly attributable to the use of latent variable modelling, which corrects for attenuation of zero-order correlations associated with measurement error⁴⁴. Importantly, efforts to capture various facets of impulsivity using a unified, purpose-built set of tasks/measures have not been reported previously. Thus, our research answers a recent call by the scientific community for the development of measures of cognitive impulsivity with sound psychometrics^{14,17}.

Criterion validity was demonstrated by the finding that, in the AMT subsamples, the monitoring/shifting factor of the CIS explained 5.8% of the variance in the addiction problems factor, reflecting covariance in self-reported alcohol use, drug use and gambling problems, and the information gathering and attentional control factors explained 12.1% of the variance in the addiction problems factor in the community and clinical sample. In combination, self-reported and cognitive measures of impulsivity explained between 26.6% and 47.0% of the variance in the addiction problems factor, with 1.3%, 2.0%, and 2.8% unique variance explained by the CIS. In contrast, a recent study¹⁶ reported 5% variance explained in-sample in daily smoking, and less than 2% variance in binge drinking, problem drinking and drug use. Cross-validation in that study yielded even lower estimates, several of which were not above chance. Additionally, the task variables did not demonstrate incremental validity when combined with the included self-report measures of impulsivity. Notably, those authors used a much bigger test battery and factor scores derived from a much larger number of dependent variables¹⁶. The CIS can therefore assist in providing mechanistic insights into the cognitive underpinnings of established impulsivity traits that have traditionally only been measured by self-report, as well as explain unique variance in real-world problem behaviours, challenging the proposed dichotomy between prediction (that is, self-report) and explanation (that is, cognition)¹⁸.

The meaningful benefit of the monitoring/shifting factor in explaining variance in the addictions problem factor, even when the five facets of self-reported trait impulsivity were included in the model, is notable given the large amount of empirical overlap between the UPPS-P, Alcohol Use Disorders Identification Test (AUDIT), Drug Use Disorders Identification Test (DUDIT) and Problem Gambling Severity Index (PGSI) due to informant bias

and common method variance⁴⁰. Furthermore, different cognitive mechanisms seem to explain different levels of problem severity, as suggested by the unique contribution of the information gathering factor to addiction problems in the sample including help-seeking participants. Different mechanisms for different severities may contribute to explain deviation from our original hypothesis. The prospector's gamble task measures reward valuation aspects of impulsivity, which have been consistently associated with addiction vulnerability⁴⁵. Thus, the monitoring/shifting factor may be capturing aspects of disinhibition associated with general vulnerability to addiction, hence it is not surprising that it was the only factor associated with addiction problems in the community-based AMT sample. In contrast, the finding that information gathering (but not monitoring/shifting) was associated with addiction problems in the clinically enriched replication sample aligns with evidence that sensation-seeking tendencies are a specific marker of addiction maintenance in endophenotype studies⁴⁶. In line with this interpretation, we found that information gathering was more consistently associated with sensation seeking compared with the other four self-reported facets of impulsivity, suggesting that it provides a reasonably specific index of the cognitive mechanism underpinning this trait. These results imply that the CIS is, in principle, well positioned to address some of the recently highlighted limitations of cognitive tasks of impulsivity versus self-reports¹⁶ and may be used to study cognitive processes germane to substance and behavioural addictions, although more research is needed to consolidate these findings.

The consistency of performance across two different administrations and the ability to detect between-subjects variance speak to the reliability of the CIS, as well as its applicability for pre-post controlled studies and individual differences research. All (except one) performance measure showed substantial test-retest reliability. We can gain a better understanding of our reliability results by considering the recent work of Enkavi et al.¹⁷, in which a large-scale examination of test-retest reliability was conducted in a battery of self-report and behavioural task measures of self-regulation, an overarching construct encompassing impulsivity. Their findings (which have been mirrored elsewhere³²) demonstrate that, on average, the test-retest reliability of behavioural tasks is generally low, that is, a median ICC of 0.31. This is purportedly because behavioural tasks suffer from low between-subjects variance (and an insensitivity to individual differences)^{17,32}. By contrast, our results indicate that, for almost all dependent variables, the CIS performs well above average in terms of test-retest reliability and between-subjects variance, compared with other cognitive tasks. Although the fast identification errors measure from the caravan spotter task showed only moderate reliability, it was still within the range described for similar tasks. Additionally, incorporating multiple task performance variables as indicators within a latent variable model captures only common variance in the construct of interest, such that task variance attributable to unreliability is excluded from the model⁴⁴.

A potential additional reason for the higher reliability of the CIS is the use of accuracy as a metric of performance rather than reaction time or difference scores. Difference scores have routinely been used to quantify performance on traditional tasks used to measure impulsivity. However, difference scores have been shown to have poor reliability, which decreases as a function of the reliability of performance on task conditions and the correlation between them³¹. For this reason, researchers have argued against the continued use of difference scores and, in particular, traditional tasks used in experimental psychology^{31–33}. In addition to these factors, it is possible that the shorter retest delay (1 week), compared with prior studies^{17,28,32} (3 weeks to 6 months) contributed to better reliability, which is higher across shorter periods⁴⁷. Future studies should examine the test-retest reliability of the CIS over longer delays, although the 1-week interval is standard in validation

studies of cognitive testing⁴⁸, and similar intervals are frequently used in experimental studies and drug trials⁴⁹, and becoming more relevant in the context of ecological momentary assessments. Thus, our results have direct implications for the applicability of the CIS in multiple contexts.

Finally, the dependent variables of the CIS did not significantly differ between the face-to-face and online samples. This latter finding supports the validity of using large-scale, online testing to administer behavioural laboratory measures of disinhibition. It may be that our use of gamification elements promoted engagement with the cognitive tasks delivered online given the high retention rates in both of our samples and as revealed by qualitative feedback. Subjective feedback from participants showed they found the tasks enjoyable and engaging, and thought they had an appropriate level of difficulty. Instructions were deemed clear, with the only exception of the prospector's gamble task. The latter finding is not unexpected, as instructions are intentionally ambiguous and participants are required to learn via feedback during the task. Participants also found the tasks relatively long, which is something that we are currently addressing via item response theory modelling, so that we can abbreviate the CIS to increase end-users' uptake while maintaining psychometric strength.

The study has potential limitations. The difference in parameter estimates between subsamples might be at least partly attributable to missing data. Although the proportion of missing data for the CIS task variables was relatively small (5.9–9.3%), it was attributable to an indirect missing not at random (MNAR) mechanism, which can introduce bias into the parameter estimates⁵⁰. We were able to demonstrate through sensitivity analyses that bias was relatively small for most parameters, and full-information maximum likelihood and multiple imputation converged on very similar estimates. Nevertheless, the fraction of missing information was high for the standardized loading estimates of the attentional control factor, particularly in the AMT validation subsample, which can introduce unstable estimates. The attentional control factor was stable across the AMT calibration subsample and the community and clinical sample, and the information gathering and monitoring/shifting factors exhibited very stable psychometric properties across all three of the samples. This further suggests that the observed instability in the attentional control factor across samples was largely attributable to missing data, although it is important to acknowledge that similar constructs (that is, interference control/inhibition) have been shown to be difficult to measure in a latent variable context and conceptually disputed^{29,43}. The lack of incremental criterion validity of the attentional control factor could be due to the limited sensitivity of behavioural (versus, for example, neuroimaging) indices from attention-related inhibition tasks to predict addiction problems⁴⁵ and/or lack of meaningful association between the specific mechanisms captured by the bounty hunter task and addictive behaviours.

In terms of convergent validity, it is possible that the findings are limited to the specific cognitive impulsivity tasks chosen for comparison (that is, SART, JTC and IGT). Regarding criterion validity, it is worth noting that our study was cross-sectional and, thus, provides initial information regarding correlations between cognitive processes driving impulsivity and real-world behaviours. We provide evidence of clinical validity of the CIS which is limited to substance and behavioural addictions. Future studies should examine whether the CIS is a valid predictor of other disorders such as ADHD, bipolar and personality disorders. High reliability estimates suggest that the CIS can be used to reliably assess cognitive processes underpinning disinhibition. Computational models linking accuracy measures with response time distributions and feedback schemes could be used in the future to tease apart subcomponent cognitive processes and brain underpinnings of each task⁵¹. Future studies could use the CIS in both laboratory and online studies, and

apply it to improve understanding of individual differences and clinical populations.

Methods

Ethics statement. Our research complies with all relevant ethical regulations. The Monash University Human Research Ethics Committee approved the study (MUHREC project numbers 7,536 and 11,999), and all participants signed an informed consent form. Participants included in the online AMT sample (494 female, 486 male, 3 nonbinary/undisclosed, 73 missing) were reimbursed US\$6 USD based on ethical payment rate guidelines for MTurk studies. Participants in the community and clinical sample (285 females, 281 females, 12 nonbinary/undisclosed) were reimbursed AUD\$60 gift cards. Participants in the face-to-face study (42 females, 20 males, 1 undisclosed) were reimbursed AUD\$20 for completing the two testing sessions.

Participants. *Online AMT sample.* Participants were 1,056 adults living in the United States and aged 18–77 years ($M = 36.85$ years, $s.d. = 10.6$ years) who were recruited from MTurk (www.mturk.com) (see also Supplementary Methods). Participants were included if they were over 18 years of age, not accessing the link via a mobile device, had no colour blindness/deficiency and had not been diagnosed with a neurological illness. The sample was randomly split into calibration ($n = 528$) and validation ($n = 528$) subsamples for the purposes of model cross-validation using invariance testing. The age ($t(958) = -1.737$ ($\Delta M = -1.187$, $s.e. 0.684$, 95% CI -2.529 to 0.154), $P = 0.083$) and education ($t(917) = -0.574$ ($\Delta M = -0.087$, $s.e. 0.151$, 95% CI -0.383 to 0.210), $P = 0.566$) of participants were comparable between the subsamples.

Community and clinical sample. Community participants were 319 individuals (112 male, 203 female, 4 nonbinary/undisclosed) aged 18–65 years ($M = 30.49$ years, $s.d. = 9.72$ years) recruited from a participant database for the Monash Brain and Behaviour Project and through online advertisements via the project Facebook page. Clinical participants were 259 (169 male, 82 female, 8 nonbinary/undisclosed) help-seeking individuals aged 18–61 years ($M = 25.59$ years, $s.d. 8.11$ years) recruited from Counselling Online, an Australian national online treatment web portal for addiction problems. The selection criteria and screening process are detailed in the pre-registration protocol (<https://osf.io/vhn95/>).

Face-to-face testing sample. Sixty-three adults aged 18–45 years ($M = 24.17$ years, $s.d. = 6.02$ years) were recruited through both online (for example, SONA, Facebook) and physical (for example, flyers posted in the community) advertisements (see also Supplementary Methods). Participants were included if they had no history of head trauma/neurological illness, were not currently diagnosed with a psychiatric disorder and had English as a first language. The demographic differences between the face-to-face sample and the other samples reflect the different strategies of recruitment: the online samples via MTurk/Counselling Online, and the lab sample via the Turner Institute's Sona Systems online platform for research participants management and community and social media advertisements that tend to attract younger and more educated participants.

Measures. *CIS.* The CIS comprises three individual tasks, each measuring one of the cognitive drivers of disinhibition: (1) bounty hunter (attentional control), (2) caravan spotter (information gathering) and (3) prospector's gamble (feedback monitoring/shifting). The three tasks have a common theme and layout (the 'Wild West'), and similar format and structure (similar visual designs, four-block structure with 40–60 trials each, chosen based on the length of similar well-validated measures³², and the same type of response required, that is, binary choices). At the beginning of the CIS, participants are given a goal, namely, 'Reach your brother's newly purchased and run-down ranch with enough cash to fund renovations and fixes'. During the CIS, the respondent is stopped three times to complete one of the three individual tasks and earn in-game money. A points-based system is employed to incentivize participants to choose quickly between alternatives, with the aim of obtaining as much in-game money as possible. The order of the three tasks is randomized by default, but it can be pre-determined. The CIS takes approximately 45 min to complete with little variation.

Bounty hunter. This task was designed to measure attentional control. On each trial, participants are presented with one of two stimuli, namely a bandit or a sheriff character. The goal of the task is to shoot the bandits ('go') but not the sheriffs ('no-go'). Participants are instructed to make quick and accurate 'go' (press spacebar) or 'no-go' (withhold from pressing spacebar) responses. They are also instructed to attend to cue signals based on the time of day: dawn ('no-go') and dusk ('go'), which is similar to a cued go/no-go task³². The task consists of four blocks of 60 trials each (240 in total), with new iterations of the characters presented in each block. Stimulus onset asynchronies (SOAs, that is, the interval between the cue and the stimulus) are manipulated such that some SOAs are very short (50 ms, 200 ms) and some are much longer (1,200 ms, 3,000 ms). This manipulation challenges mobilization (short SOAs) and maintenance (long SOAs) of attentional resources, as both can trigger impulsive behaviours⁵². The four types of trials (with SOAs of 50, 200, 1,200 and 3,000 ms) are presented in a

randomized order at an approximately equal rate across the four blocks. On each trial, participants can earn or lose US\$50, which is delivered as correct or incorrect feedback. Participants are provided with a 1000 ms response window, and the inter-trial fixation stimulus is 800 ms. The key performance indices of the task are the proportion of correct responses and commission errors (CEs; that is, responses on no-go trials), which are measured overall as well as at short (50 ms, 200 ms) and long (1,200 ms, 3,000 ms) SOAs.

Caravan spotter. This task was designed to measure information gathering during speeded decisions. The goal of this task is to safely guide a trade caravan through the desert by spotting dangers and obstacles. On each trial, participants are presented with a totally ambiguous pixelated image that gradually disambiguates over the course of the trial. Images initially appear 50% pixelated and linearly transition to 100% unpixelated across the response window of 2,000 ms. The images are presented in a circle, which represents a telescope to see hazards in the distance. Participants are instructed to decide which of two stimuli the pixelated image will ultimately morph into (for example, a wagon or a horse). The task consists of four blocks of 60 trials each (240 in total), with new stimuli categories presented each block. Participants are awarded points (in-game money) for speed and accuracy, but penalized for incorrect responses. Specifically, they are awarded 1–200 points for correct responses (10 points lost every 100 ms), given a –100 penalty for incorrect responses and 0 points for no response. This manipulation fosters impulsive errors in individuals characterized by ‘low reflection’, as they tend to have a lower requirement for information gathering before making a decision. The key performance indices of the task are the proportion of correct responses and fast identification errors (that is, the number of times an incorrect response was made in <1,000 ms).

Prospector’s gamble. This task was designed to measure feedback monitoring/shifting, by modifying previously described probabilistic reversal tasks¹⁵. On each trial, participants are presented with two characters (gold prospectors). The goal of the task is to select the prospector who can get them the greater amount of gold, considering that one of them will always be luckier than the other one, but also that their luck may change. Participants are instructed to choose the luckier of the two prospectors to work with every day. The task comprises four blocks of 40 trials each (160 in total), with a contingency reversal occurring between each block. Incorrect feedback is provided on 20% of trials in blocks 1 and 2, and on 30% of trials in blocks 3 and 4. The contingency reversals and block changes are not revealed, thus over the course of the trials, participants must learn which stimulus is ‘correct’ and continuously select it, despite the correct stimulus infrequently giving incorrect feedback, and vice versa.

The correct/incorrect feedback delivered after each trial is manipulated, such that it has a 50% chance of being delivered in a highly stimulating and engaging format (that is, with animations, sounds and more extreme outcomes) and, thus, providing a greater magnitude of reward or punishment, in contrast to the standard feedback. In the standard feedback condition, US\$50 is awarded for ‘correct’ decisions and US\$0 is penalized for ‘incorrect’ decisions. In the enhanced feedback condition, the outcomes were US\$200 and –US\$100, respectively. This manipulation of the intensity of positive and negative feedback, that is, standard versus enhanced, enabled us to examine whether participants make more impulsive errors following more intense positive or negative feedback (akin to the concept of positive and negative urgency). To foster fast (impulsive) responses following positive or negative feedback, we included a time pressure element such that participants need to make selections in a short (1,000 ms) interval. The key performance indices of the task are the proportion of correct responses and total perseverative errors (that is, the number of trials a participant selects the previously correct stimulus following stimulus change), which are calculated overall and separately for trials following positive and negative feedback for both basic and enhanced versions.

Convergent/construct validity measures. The SART has been widely used to measure attentional contributions to response inhibition²⁴. The beads task, together with the information sampling task, are the only computerized tasks that measure information gathering²⁵, and we chose the first for practical reasons; it had no financial cost. The IGT was chosen because it measures both feedback monitoring and shifting processes²⁶, is sensitive to impulsive behaviours⁴ and has been used to validate some of the underlying neural mechanisms of feedback-related contributions to impulsive behaviour.

SART. This computerized go/no-go task measures sustained attention and response inhibition. In the SART, single-digit numbers are presented in a randomized order every 1,150 ms. Participants are required to withhold their response to a single no-go number (non-target stimuli) that occurs in 11% of cases and respond to all other numbers (go-target stimuli). Due to its steady, regular pacing and the infrequency of targets, the SART fosters fast, automatic responding. The key performance indicator is no-go success (suppression of responses on no-go trials). The SART has adequate reliability and validity²⁴.

JTC. The JTC beads task is used to assess reflection impulsivity²³. Participants are presented with two jars (A and B) containing different ratios of red to blue beads

(85:15 or 60:40). Participants are told that the computer selects one of the jars and beads are drawn at random from it. The computer draws one bead at a time and always returns the bead to the jar before it draws the next bead, so that the ratio stays the same. The participant has to decide which jar (A or B) the computer is drawing the beads from. The design is consistent with the ‘draws to decision’ methodology as participants can request as many beads as they need to before making a decision. A sequence of red and blue beads at the bottom of the screen reminds the participant of all the beads previously drawn. The task comprises four blocks with three trials in each. Two blocks present jars with probabilities 85:15 and two blocks present jars with probabilities 60:40 (the order being randomly determined). The proportions of the two bead colours were counterbalanced across participants. The key performance index is the median number of beads drawn before a decision is made (summed across blocks with probabilities 85:15 and 60:40, respectively).

IGT. This is a decision-making task that measures the ability to learn from negative and positive feedback (possible monetary gains/losses) in the context of uncertainty and risk. Participants are required to complete 100 trials in which they choose from one of four decks of cards. Decks A and B are considered ‘risky decks’ as they generate high rewards (for example, US\$100 per win), but also high losses, yielding an overall negative tally. Decks C and D are considered ‘safe decks’ as they generate smaller gains (for example, only US\$50 per trial), but also small losses, yielding an overall profit. The key performance index is the participant’s net score, calculated by subtracting the number of ‘safe deck’ choices from ‘risky deck’ choices. The IGT has good reliability and validity in relation to real-world impulsive behaviours²⁴.

Short UPPS-P impulsive behaviour scale. This shortened version of the UPPS-P impulsive behavioural scale²⁷ comprises five subscales (four items per scale) that each measure a distinct personality trait that leads to impulsive behaviour. The subscales are: negative urgency (tendency to act rashly when experiencing strong negative emotions), lack of perseverance (inability to remain focused on a task), lack of premeditation (tendency to act without thinking), sensation seeking (tendency to seek out novel and thrilling experiences) and positive urgency (tendency to act rashly when experiencing strong positive emotions). Each item is scored on a four-point Likert scale ranging from 1 (strongly agree) to 4 (strongly disagree). Total scores are summed for each scale, with higher scores indicating increased impulsivity. The short UPPS-P has good internal reliability ($\alpha = 0.65–0.88$) and criterion validity as the subscales correlate with measures of substance use²⁴.

Clinical validity measures. AUDIT. This ten-item self-report tool assesses alcohol use and related consequences over the past year. Each item is rated on a five-point Likert scale. Total scores range from 0 to 40, with higher scores indicative of increased alcohol use. AUDIT has excellent internal consistency ($\alpha = 0.80$) and test–retest reliability, and is a valid indicator of alcohol use severity²⁵.

DUDIT. This 11-item self-report scale assesses illicit drug use and related consequences over the past year. Each item is rated on a five-point Likert scale. Total scores range from 0 to 44, with higher scores indicative of increased drug use. The DUDIT has excellent internal consistency ($\alpha = 0.80–0.97$) and test–retest reliability ($\alpha = 0.77$ to 0.91), and is a valid indicator of drug-use-related problems²⁶.

PGSI. This nine-item self-report scale assesses gambling behaviour and associated problems over the past year. Each item is rated on a four-point Likert scale. Total scores range from 0 to 27, with higher scores indicative of increased gambling behaviour. The PGSI has acceptable internal consistency ($\alpha = 0.63$) and test–retest reliability ($\alpha = 0.78$) and is a valid indicator of gambling behaviour in clinical and non-clinical populations²⁷.

Procedure. Online data collection. The testing of the original community sample was facilitated by TurkPrime, an online crowdsourcing platform that integrates with the Amazon MTurk platform, which we recruited the participants from. Participants were first directed to a survey hosted by Qualtrics where they were provided with a digital explanatory statement and gave informed consent. Second, they were redirected from Qualtrics to complete the CIS, then redirected back to Qualtrics to complete a demographics survey and self-report measures of alcohol use, drug use and gambling (AUDIT, DUDIT and PGSI). Participants completed a demo version of the CIS before starting the tasks to minimize technical errors and performance anxiety. Finally, participants were directed to Inquisit, a study platform hosted by Millisecond (www.millisecond.com), where they first completed a self-report measure of trait impulsivity (short UPPS-P) before being randomly allocated to complete one of three cognitive tasks (JTC, IGT or SART). They were notified that they would be required to install the Inquisit launcher (and would be redirected to a page with installation instructions) during the informed consent process. The testing of the replication community and clinical samples followed the same procedures, although instead of MTurk we used the Counselling Online web portal (<https://www.counsellingonline.org.au/>) for the clinical sample, and dedicated web links for the second community sample. Assessments were the

same, although they did not conduct the 'validation' cognitive tasks (JTC, IGT or SART) to reduce participant burden.

In-person data collection. This study used a repeated-measures, within-subjects design to assess the reliability of CIS performance indices across two sessions spaced 1 week apart, which were conducted in a face-to-face laboratory setting. Participants completed two testing sessions at the same time of day in a 1-week interval. On both testing session days, participants were instructed to adhere to their usual level of caffeine intake, and to refrain completely from consuming alcohol and any other non-prescribed drugs until after the session. To minimize practice effects due to novelty, we pre-exposed participants to a self-contained demonstration version of the CIS, which served to familiarize them with the task goals, environment and response requirements. The order of the three tasks was counterbalanced between and within subjects. Sixty (95.2%) participants completed both sessions. A subgroup of participants ($n = 12$) completed a slightly modified version of the CIS to increase task engagement after initial feedback from participants (for example, we removed easily identifiable stimuli features in the bounty hunter task and randomized blocks within the caravan spotter task).

Data analyses. Rationale. Internal consistency and validity were initially tested within an online sample drawn from the Amazon Mechanical Turk community (AMT sample $N = 1,056$) and randomly split into two subsamples for cross-validation purposes (AMT calibration subsample, $n = 528$; AMT validation subsample, $n = 528$). One of three tasks of the same family (SART (attentional control), JTC (information gathering) and IGT (monitoring/shifting)) was administered to a subset of AMT participants ($n = 297$, $n = 308$, $n = 302$). All participants also completed the short form of the UPPS-P (sUPPS-P), the gold-standard measure of self-reported trait impulsivity, and self-reported measures of addiction-related problems (impulsivity-related 'real-world' outcomes), using well-validated measures of alcohol use problems (AUDIT), drug use problems (DUDIT) and gambling problems (PGSI). We also sought to replicate and extend the major findings from our discovery phase to an independent sample ($N = 578$), recruited online from the community ($n = 319$) and enriched with a clinical cohort of help-seeking individuals ($n = 259$) from a drug, alcohol and gambling problems treatment service. Test-retest reliability was examined in a laboratory study ($n = 63$), across two face-to-face sessions separated by 1 week. Since the CIS was designed to be administered both online and face-to-face, we also tested the equivalence between these two administration modalities.

We used gold-standard structural equation modelling (SEM) techniques to evaluate the reliability and validity of the CIS. First, we evaluated the factorial structure of the CIS and the discriminant validity of the attentional control, information gathering and monitoring/shifting constructs using a 'competing models' approach⁵⁸. This is a powerful and rigorous approach to determining the best-fitting model based on direct statistical comparison of several theoretically plausible alternative 'measurement' models using CFA. We then examined the reliability of measurement of these factors by calculating construct reliability⁵⁹ and their discriminant validity⁶⁰. Second, we evaluated the convergent/divergent validity and specificity of the attentional control, information gathering and monitoring/shifting constructs by examining their patterns of linear associations with the five self-reported facets of impulsivity represented by the sUPPS-P within a CFA model. This approach allows the empirical association between constructs to be estimated as factor intercorrelations, which corrects for measurement error⁶¹. Third, we evaluated the criterion validity of the CIS within a structural regression model, by incorporating the AUDIT, DUDIT and PGSI variables into the model as indicators of a unitary addiction problems factor. These analyses were conducted to determine whether the attentional control, information gathering, and monitoring/shifting factors explained statistically and practically significant variance in a common underlying vulnerability to addiction problems, as recent research has questioned the ability of cognitive tasks (versus surveys) to predict real-world outcomes¹⁶. Fourth, we incorporated the five self-reported facets of impulsivity represented in the sUPPS-P as additional factors in the structural regression model. This approach enabled us to test and quantify the incremental criterion validity of the CIS in explaining variance in real-world behaviours in addition to variance explained by self-reported impulsivity. For each of these analyses, we first estimated the models in the AMT calibration subsample and then sought to replicate the dimensional structure, factor loadings, intercepts/means, indicator error variances, factor variances and covariances, and structural regression paths in the AMT validation subsample and independent community and clinical sample using invariance testing^{61,62}. Invariance testing is a rigorous and powerful statistical approach to evaluating model reproducibility that involves imposing a series of increasingly restrictive equality constraints on the model parameter estimates^{61,62}. Equivalence (that is, invariance) of the unstandardized estimates for the different model parameters provides a formal statistical test of model stability and replicability. Finally, test-retest reliability was tested across two laboratory sessions, and we assessed the equivalence of performance across online and face-to-face CIS administrations.

Structural equation modelling. Model specification and estimation. SEM was conducted in Mplus 8.3 using the covariance matrix⁶³. Models were estimated

using full-information maximum likelihood with the expectation-maximization algorithm to account for missing data and the Bollen–Stine bootstrap procedure with 10,000 posterior draws to account for multivariate non-normality⁶³. This approach uses resampling with replacement to generate a pseudo-population that does not have attenuated standard errors associated with the leptokurtic distributions often observed with performance-based task data⁵⁰. Bootstrapped standard errors and 95% CIs, reported in the text and in model figures, provide an indication of the precision of the model parameter estimates that are less biased by multivariate non-normality⁶⁰. Mplus defaults for the number of iterations (1,000) and convergence criterion (0.000005) were initially used for model estimation⁶³. The number of iterations was increased to 50,000 for the invariance model of the CIS using the effects coding method. This was to overcome convergence problems encountered using the default number of iterations, probably associated with the multi-group linear constraints on model parameters used for factor scaling with the effects coding method (see below).

Latent variables are unobserved, dimensionless variables. The scale of latent variables must be estimated as part of the model identification process. We used the effects coding method to set the scale of each latent variable for the CIS factors³⁶. This method constrains the variance of each latent variable to be the average explained variance across all indicators weighted by their loading on that factor. Effects coding is the preferred method when conducting multi-group analyses with invariance testing because it does not require assumptions of between-group equivalence, such that either the factor variance or loading of the first variable (that is, reference variable) are fixed to equality. The reference variable method was used to scale factors for the remainder of the models.

Simple factor structure was tested for each of the hypothesized models, such that observed variables (that is, indicators) were specified as loading only on their target factors (that is, non-target cross-loadings constrained to zero). The assumptions of conditional independence and independent clusters (that is, that all observed variables load on only one factor and are independent apart from their target factor loadings) impose unrealistic assumptions on models of complex data, particularly performance on cognitive tasks with shared surface characteristics and common method variance. Therefore, we made post hoc model changes by freely estimating covariances between unexplained error variances in the observed indicators where theoretically plausible and empirically indicated by correlation residuals, standardized covariance residuals and modification indices⁶⁴. Post hoc model changes capitalize on chance, thus we corrected for multiple comparisons using the B–H false discovery rate (FDR; $q = 0.05$)⁶⁵.

Evaluating model fit. Overall model fit is evaluated with 'global' fit indices based on the discrepancy between the observed covariance matrix (linear relationships in the sample data) and the model reproduced covariance matrix (linear relationships implied by the freely estimated parameters and constraints specified in the hypothesized model). The chi-squared (χ^2) test statistic is the gold-standard index of global fit, with higher probability values signifying greater probability of the null hypothesis of exact fit of the model reproduced covariance matrix to the observed covariance matrix^{66,67}. We also report three 'approximate fit indices': the RMSEA and associated 90% CI, the CFI and the SRMR (refs. ^{39,68}). Higher values for the CFI and lower values of the RMSEA and SRMR are indicative of better-fitting models⁶⁹. Local fit of model parameters was evaluated with reference to the strength of point estimates for the path coefficients (higher estimates indicating stronger linear relationships), as well as the size of bootstrapped s.e. and 95% CIs (smaller s.e. and 95% CI indicate more precise estimates). We also inspected and report where appropriate the correlation residuals (ϵ), which represent absolute differences between expected and model reproduced correlations. A pattern of correlation residuals exceeding the recommended threshold of $\epsilon > 0.10$ may be indicative of poor local fit⁶⁹.

Models with the same latent variables but different structural parameters (that is, factor intercorrelations or regression paths between factors) freely estimated or constrained to zero or some other value are statistically 'nested'. The difference in model fit between these models as given by $-2 \times \log$ likelihood is χ^2 distributed and can be tested using the χ^2 difference test ($\Delta\chi^2$) with associated P value⁶⁶. Statistically non-nested models were compared using the Bayesian information criterion (BIC). Lower BIC values indicate a more parsimonious and better-fitting model. To facilitate model testing, we used Bayesian statistics, specifically the Bayesian conditional posterior probability [$\text{Pr}_{\text{BIC}}(H_i|D)$]. The Bayesian conditional posterior probability is derived from the BIC, which takes into account model complexity by incorporating the number of estimated parameters (k) in the formula $-2 \ln \hat{L}_i + k_i \ln(n)$, where $2 \ln \hat{L}_i$ is the log likelihood estimate for model i and n is the number of observations⁷⁰. The Bayesian conditional posterior probability quantifies the relative probability ($P = 0.00\text{--}1.0$) of model i compared with k models by dividing the exponentiated $-\frac{1}{2}\text{BIC}$ for model i by the sum of the exponentiated $-\frac{1}{2}\text{BIC}$ for k models: $\text{Pr}_{\text{BIC}}(H_i|D) = \frac{\exp[-\frac{1}{2}\text{BIC}(H_i)]}{\sum_{j=1}^k \exp[-\frac{1}{2}\text{BIC}(H_j)]}$ (ref. ⁷⁰).

Modelling strategy. Complex models with large numbers of empirically related variables are prone to convergence problems (that is, inability of the modelling software to converge on admissible solutions) and misspecification (specification of model parameters that are not consistent with the underlying linear relationships/data generation process). To circumvent these problems, we used a jigsaw-piece

modelling strategy, determining the best-fitting model for each set of constructs separately before incorporating them into combined models⁷¹. This approach enabled us to find the optimal representation of each set of constructs and their interrelationships and informed correct specification of the regression parameters for subsequent models. We compared several theoretically plausible alternative ‘measurement’ (that is, CFA) models using a competing-models approach before determining the optimal latent structure of each set of constructs in the AMT calibration subsample. Each of the final models selected based on global and local fit testing was cross-validated in the AMT validation subsample and the independent community and clinical sample using invariance testing⁶². We used a free-baseline approach which entails fewer assumptions and proceeds from testing the least restrictive to increasingly strict equality constraints on model parameters⁴⁴. We tested for invariance in the following order: (1) configural invariance (same pattern of factor loadings), (2) weak invariance (statistical equivalence of unstandardized factor loading estimates), (3) strong invariance (statistical equivalence of intercepts), (4) equality of latent means, (5) strict invariance (equality of unstandardized error variances and covariances) and (6) structural invariance (equality of factor variances and covariances)⁶¹. Partial invariance was considered where full invariance did not apply⁶⁸. We used a backwards sequential search strategy⁷², by freeing parameters for estimation one at a time in the AMT validation subsample and community and clinical sample based on inspection of between-group differences in unstandardized estimates. The backwards sequential search method has been shown to have good sensitivity for detecting true positives while also yielding a low rate of false positives (that is, type I error control)⁷³. Invariance was tested using the $\Delta\chi^2$ test and a designated P value threshold of $P < 0.05$ (ref. ⁶⁶). The $\Delta\chi^2$ is well powered to detect differences between groups when sample size is large.

Model replicability and discriminant validity. The index H ($H = 1/[1 + \frac{1}{\sum_{i=1}^k \frac{\lambda_i^2}{1-\lambda_i^2}}]$)

was calculated to quantify the construct reliability and replicability of each of the three CIS factors⁵⁹; where λ is the fully standardized factor loading, standardized using the variances of the continuous latent variables and observed background variables⁶³. H reflects the proportion of variance in a construct explained by its indicators and varies from 0 to 1, with higher values indicating greater construct reliability and replicability⁵⁹. Higher factor loadings increase the ratio of explained to unexplained variance across the indicators of a latent variable and increase the likelihood that a latent variable is empirically well defined and will be reproducible across studies⁵⁹. Discriminant validity can be demonstrated by showing that the average variance extracted (that is, the average of the fully standardized factor loadings for the measured variables [$\frac{\sum_{i=1}^n \lambda_i^2}{n}$]) for each factor exceeds the amount of variance shared with other factors, as quantified by the squared multiple correlation (that is, the squared factor intercorrelation coefficient ϕ^2)^{60,74}, where ϕ is the fully standardized factor correlation coefficient, standardized using the variances of the continuous latent variables and observed background variables⁶³.

Interpretation of model parameter estimates. We follow proposed guidelines in abandoning strict observance of, and sole reliance on, the arbitrary P value threshold of <0.05 when interpreting parameter estimates⁷⁵. Instead, we report the results of statistical tests of individual parameters and overall model fit in terms of continuous P values, along with bootstrapped standard errors, and 95% BCIs to facilitate evaluation of the precision of statistical estimates⁷⁵. Additionally, we interpret results in the context of the null hypothesis of zero association in the population ($r=0.00$) where the model parameters reflect population effect size estimates (r_{es}), such that 0.10, 0.30 and 0.50 correspond to small, medium and large effects, respectively. However, to control for potential type I error in parameter estimation within models whilst minimizing the inflated type II error associated with Bonferroni correction, we corrected the threshold for determining their statistical significance using AB2: $\alpha_{\text{per test}} = \frac{\alpha_{\text{familywise}}}{k - \sqrt{|J|}}$; where $\bar{r}_j$ is the absolute average correlation of parameter j

with other parameters in the model and k is the number of structural parameters being evaluated for statistical significance against the null hypothesis³⁷. We used the average of the fully standardized factor intercorrelations ($\bar{\phi}$) within the model as a proxy measure of parameter dependency.

SEM notation. To refer to model parameters in the text, we use linear structural relations notation, which uses lower-case Greek letters to refer to different elements of the model⁷⁶. Variances and covariances between exogenous (independent) latent variables and endogenous (dependent) latent variables are denoted by lowercase phi (ϕ) and psi (ψ). Regression parameters of endogenous on exogenous latent variables are denoted by gamma (γ). Unexplained variance (that is, error variance) in observed variables for exogenous and endogenous latent variables, as well as covariances between them (that is, error covariances), are denoted by theta with subscript delta (θ_δ) and theta with subscript epsilon (θ_ϵ), respectively. Model figures are displayed using symbols from the McArdle–McDonald reticular action model⁷⁷. Observed (also measured or manifest) variables are represented as rectangles. Latent variables are represented as ellipses. Factors (or constructs) are represented as large ellipses. Disturbances (that is, unexplained variance in

an endogenous latent variables) and error variances for observed variables are symbolized with small ellipses. Double-headed curved arrows pointing to factors are the latent variable variances (ϕ). Single-headed straight arrows pointing from small ellipses to endogenous latent variables are the disturbances (that is, unexplained variance in a factor by the exogenous latent variables). Straight single-headed arrows from large ellipses to observed variables reflect factor loadings. Straight single-headed arrows pointing from small ellipses to measured variables are the variances (that is, error) in the variables not explained by the factor (θ_δ or θ_ϵ). Curved double-headed arrows between large ellipses are factor (that is, latent) intercorrelations (ϕ or ψ) and between small ellipses are error covariances (θ_δ or θ_ϵ). Single-headed straight arrows between factors are regression coefficients of an endogenous latent variable on an exogenous latent variable (γ).

Factor loadings for the measured variables (for example, no-go commission errors, fast response errors and perseverative errors) are specified with a negative sign, such that higher scores on the latent variables reflect better/more efficient attentional control, information gathering and monitoring/shifting.

Test–retest reliability. Estimates were derived with the ICC. We used a two-way mixed-effects model based on a single administration of the CIS and absolute agreement between the two administrations (as opposed to mere consistency). In the commonly cited Shrout and Fleiss nomenclature, this corresponds to ICC [3A,1] (ref. ⁷⁸). Because the ICC assumes normally distributed data, we transformed variables with an absolute skewness >1 using two different transformations (arcsine and folded root), depending on the characteristics of the variable distribution. For each dependent variable of interest, we report the ICC estimate and 95% confidence intervals. There are no standardized values for interpreting ICC estimates, but these (or similar) thresholds are typically used: excellent (>0.80), good/substantial (0.60–0.80), moderate (0.40–0.60) and poor (<0.40) (ref. ⁷⁹). As part of ICC analyses, we also decomposed and compared different sources of variance in the data, that is between-subjects, within-subjects and error variance³². Measures with proportionally lower between-subjects variance (compared with within-subjects variance) are able to produce robust experimental effects, but are unsuitable for analysis of individual differences and vice versa³².

Equivalence between laboratory and online administrations. To compare equivalence of performance on CIS variables between the two testing sites, namely face-to-face/ laboratory and online, we performed a series of independent-samples t tests. We used data from the first administration of the CIS in the face-to-face study and data from the online study to examine equivalence between these testing modalities. We corrected for multiple comparisons using the B–H false discovery rate ($q=0.05$) (ref. ⁶⁵).

Deviation from pre-registration. Planned factor mixture modelling analyses for determining whether the community and clinical participants could be combined into a homogeneous sample could not be completed as most of the models with two or more latent classes failed to converge or yielded untrustworthy estimates. The few models specifying two or more latent classes that did converge failed to provide a better fit to the data compared with the baseline model that treated participants from the community and clinical cohorts as a homogeneous sample, as indicated by comparison of the BIC. We therefore replaced the factor mixture modelling approach with invariance tests to establish the measurement equivalence of the CIS three-factor model and unidimensional model of addiction problems.

In hypothesis 1, we included ADHD symptoms as an additional index of ‘real-world’ behaviours, but prior to commencement of data analyses, we decided to examine and report ADHD data in a separate report.

Reporting summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data Availability

Data obtained from cognitive testing and self-report and analysed for this manuscript can be accessed via the Open Science Framework (<https://osf.io/7qrv5/>).

Code Availability

Interested researchers can gain access to the CIS by contacting the corresponding author (A.V.-G.). Mplus 8.3 code used to estimate the structural equation models presented in this manuscript can be accessed via the Open Science Framework (<https://osf.io/jqbs5/>).

Received: 4 February 2020; Accepted: 16 April 2021;
Published online: 27 May 2021

References

1. Nigg, J. T. Annual research review: on the relations among self-regulation, self-control, executive functioning, effortful control, cognitive control, impulsivity, risk-taking, and inhibition for developmental psychopathology. *J. Child Psychol. Psychiatry* **58**, 361–383 (2017).

2. Vassileva, J. & Conrod, P. J. Impulsivities and addictions: a multidimensional integrative framework informing assessment and interventions for substance use disorders. *Philos. Trans. R. Soc. Lond. B* **374**, 20180137 (2019).
3. Lima, I. M. M., Peckham, A. D. & Johnson, S. L. Cognitive deficits in bipolar disorders: implications for emotion. *Clin. Psychol. Rev.* **59**, 126–136 (2018).
4. Dawson, A. et al. Neurocognitive correlates of medication-induced addictive behaviours in Parkinson's disease: a systematic review. *Eur. Neuropsychopharmacol.* **28**, 561–578 (2018).
5. Lansdall, C. J. et al. Apathy and impulsivity in frontotemporal lobar degeneration syndromes. *Brain* **140**, 1792–1807 (2017).
6. Dir, A. L., Coskunpinar, A. & Cyders, M. A. A meta-analytic review of the relationship between adolescent risky sexual behavior and impulsivity across gender, age, and race. *Clin. Psychol. Rev.* **34**, 551–562 (2014).
7. Hutson, P. H., Balodis, I. M. & Potenza, M. N. Binge-eating disorder: clinical and therapeutic advances. *Pharmacol. Ther.* **182**, 15–27 (2018).
8. Navas, J. F. et al. Sex differences in the association between impulsivity and driving under the influence of alcohol in young adults: the specific role of sensation seeking. *Accid. Anal. Prev.* **124**, 174–179 (2019).
9. Ciobanu, L. G. et al. The prevalence and burden of mental and substance use disorders in Australia: findings from the Global Burden of Disease Study 2015. *Aust. N. Z. J. Psychiatry* **52**, 483–490 (2018).
10. Vigo, D., Thornicroft, G. & Atun, R. Estimating the true global burden of mental illness. *Lancet Psychiatry* **3**, 171–178 (2016).
11. Hamilton, K. R. et al. Rapid-response impulsivity: definitions, measurement issues, and clinical implications. *Personal. Disord. Theory Res. Treat.* **6**, 168–181 (2015).
12. Hamilton, K. R. et al. Choice impulsivity: definitions, measurement issues, and clinical implications. *Personal. Disord.* **6**, 182–198 (2015).
13. Verdejo-Garcia, A., Lawrence, A. J. & Clark, L. Impulsivity as a vulnerability marker for substance-use disorders: review of findings from high-risk research, problem gamblers and genetic association studies. *Neurosci. Biobehav. Rev.* **32**, 777–810 (2008).
14. Sharma, L., Markon, K. E. & Clark, L. A. Toward a theory of distinct types of 'impulsive' behaviors: a meta-analysis of self-report and behavioral measures. *Psychol. Bull.* **140**, 374–408 (2014).
15. Fineberg, N. A. et al. New developments in human neurocognition: clinical, genetic, and brain imaging correlates of impulsivity and compulsivity. *CNS Spectr.* **19**, 69–89 (2014).
16. Eisenberg, I. W. et al. Uncovering the structure of self-regulation through data-driven ontology discovery. *Nat. Commun.* **10**, 2319 (2019).
17. Enkavi, A. Z. et al. Large-scale analysis of test-retest reliabilities of self-regulation measures. *Proc. Natl Acad. Sci. USA* **116**, 5472–5477 (2019).
18. Enkavi, A. Z. & Poldrack, R. A. Implications of the lacking relationship between cognitive task and self-report measures for psychiatry. *Biol. Psychiatry Cogn. Neurosci. Neuroimaging* <https://doi.org/10.1016/j.bpsc.2020.06.010> (2020).
19. Dang, J., King, K. M. & Inzlicht, M. Why are self-report and behavioral measures weakly correlated? *Trends Cogn. Sci.* **24**, 267–269 (2020).
20. Friedman, N. P. & Banich, M. T. Questionnaires and task-based measures assess different aspects of self-regulation: both are needed. *Proc. Natl Acad. Sci. USA* **116**, 24396–24397 (2019).
21. Toplak, M. E., West, R. F. & Stanovich, K. E. Practitioner review: do performance-based measures and ratings of executive function assess the same construct? *J. Child Psychol. Psychiatry* **54**, 131–143 (2013).
22. Schluter, M. G., Kim, H. S. & Hodgins, D. C. Obtaining quality data using behavioral measures of impulsivity in gambling research with Amazon's Mechanical Turk. *J. Behav. Addict.* **7**, 1122–1131 (2018).
23. Egner, T. *The Wiley Handbook of Cognitive Control* (Wiley, 2017).
24. Smilek, D., Carriere, J. S. & Cheyne, J. A. Failures of sustained attention in life, lab, and brain: ecological validity of the SART. *Neuropsychologia* **48**, 2564–2570 (2010).
25. Voon, V. Models of impulsivity with a focus on waiting impulsivity: translational potential for neuropsychiatric disorders. *Curr. Addict. Rep.* **1**, 281–288 (2014).
26. Fellows, L. K. The role of orbitofrontal cortex in decision making - a component process account. *Link. Affect Action. Crit. Contribut. Orbitofrontal Cortex* **1121**, 421–430 (2007).
27. Stuss, D. T. & Alexander, M. P. Is there a dysexecutive syndrome? *Philos. Trans. R. Soc. Lond. B* **362**, 901–915 (2007).
28. Frey, R., Pedroni, A., Mata, R., Rieskamp, J. & Hertwig, R. Risk preference shares the psychometric structure of major psychological traits. *Sci. Adv.* **3**, e1701381 (2017).
29. Rey-Mermet, A., Gade, M. & Oberauer, K. Should we stop thinking about inhibition? Searching for individual and age differences in inhibition ability. *J. Exp. Psychol. Learn. Mem. Cogn.* **44**, 501–526 (2018).
30. Gomez, P., Perea, M. & Ratcliff, R. A model of the go/no-go task. *J. Exp. Psychol. Gen.* **136**, 389–413 (2007).
31. Draheim, C., Mashburn, C. A., Martin, J. D. & Engle, R. W. Reaction time in differential and developmental research: a review and commentary on the problems and alternatives. *Psychol. Bull.* **145**, 508–535 (2019).
32. Hedge, C., Powell, G. & Sumner, P. The reliability paradox: why robust cognitive tasks do not produce reliable individual differences. *Behav. Res. Methods* **50**, 1166–1186 (2018).
33. Paap, K. R., Anders-Jefferson, R., Zimiga, B., Mason, L. & Mikulinsky, R. Interference scores have inadequate concurrent and convergent validity: should we stop using the flanker, Simon, and spatial Stroop tasks? *Cogn. Res. Princ. Implic.* **5**, 7 (2020).
34. Cyders, M. A., Littlefield, A. K., Coffey, S. & Karyadi, K. A. Examination of a short English version of the UPPS-P impulsive behavior scale. *Addict. Behav.* **39**, 1372–1376 (2014).
35. Cyders, M. A. & Coskunpinar, A. Measurement of constructs using self-report and behavioral lab tasks: is there overlap in nomothetic span and construct representation for impulsivity? *Clin. Psychol. Rev.* **31**, 965–982 (2011).
36. Little, T. D., Slegers, D. W. & Card, N. A. A non-arbitrary method of identifying and scaling latent variables in SEM and MACS models. *Struct. Equ. Modeling* **13**, 59–72 (2006).
37. Smith, C. E. & Cribbie, R. A. Multiplicity control in structural equation modeling: incorporating parameter dependencies. *Struct. Equ. Modeling* **20**, 79–85 (2013).
38. Hayduk, L. A. & Littvay, L. Should researchers use single indicators, best indicators, or multiple indicators in structural equation models? *BMC Med. Res. Methodol.* **12**, 159 (2012).
39. Hair, J. F., Black, W. C., Babin, B. J. & Anderson, R. E. *Multivariate Data Analysis* 7th edn (Pearson Education, 2014).
40. Podsakoff, P. M., MacKenzie, S. B. & Podsakoff, N. P. Sources of method bias in social science research and recommendations on how to control it. *Annu. Rev. Psychol.* **63**, 539–569 (2012).
41. Cohen, J. A power primer. *Psychol. Bull.* **112**, 155–159 (1992).
42. Rosenthal, R. & DiMatteo, M. R. Meta-analysis: recent developments in quantitative methods for literature reviews. *Annu. Rev. Psychol.* **52**, 59–82 (2001).
43. Friedman, N. P. & Miyake, A. The relations among inhibition and interference control functions: a latent-variable analysis. *J. Exp. Psychol. Gen.* **133**, 101–135 (2004).
44. Bollen, K. A. & Noble, M. D. Structural equation models and the quantification of behavior. *Proc. Natl Acad. Sci. USA* **108**, 15639–15646 (2011).
45. Verdejo-Garcia, A. & Albein-Urios, N. Impulsivity traits and neurocognitive mechanisms conferring vulnerability to substance use disorders. *Neuropharmacology* **183**, 108402 (2021).
46. Ersche, K. D., Turtton, A. J., Pradhan, S., Bullmore, E. T. & Robbins, T. W. Drug addiction endophenotypes: impulsive versus sensation-seeking personality traits. *Biol. Psychiatry* **68**, 770–773 (2010).
47. Scharfen, J., Jansen, K. & Holling, H. Retest effects in working memory capacity tests: a meta-analysis. *Psychon. Bull. Rev.* **25**, 2175–2199 (2018).
48. Vincent, A. S., Fuenzalida, E., Beneda-Bender, M., Bryant, D. J. & Peters, E. Neurocognitive assessment on a tablet device: test-retest reliability and practice effects of ANAM Mobile. *Appl. Neuropsychol. Adult* **28**, 363–371 (2021).
49. Caddy, C. et al. Ketamine and other glutamate receptor modulators for depression in adults. *Cochrane Database Syst. Rev.* <https://doi.org/10.1002/14651858.CD011612.pub2> (2015).
50. Enders, C. K. *Applied Missing Data Analysis* (Guilford Press, 2010).
51. Peters, J. & D'Esposito, M. The drift diffusion model as the choice rule in inter-temporal and risky choice: a case study in medial orbitofrontal cortex lesion patients and controls. *PLoS Comput. Biol.* **16**, e1007615 (2020).
52. Pike, E., Marks, K. R., Stoops, W. W. & Rush, C. R. Cocaine-related stimuli impair inhibitory control in cocaine users following short stimulus onset asynchronies. *Addiction* **110**, 1281–1286 (2015).
53. Banca, P. et al. Reflection impulsivity in binge drinking: behavioural and volumetric correlates. *Addict. Biol.* **21**, 504–515 (2016).
54. Verdejo-Garcia, A., Bechara, A., Recknor, E. C. & Perez-Garcia, M. Decision-making and the Iowa Gambling Task: ecological validity in individuals with substance dependence. *Psychol. Belg.* **46**, 55–78 (2006).
55. Lundin, A., Hallgren, M., Balliu, N. & Forsell, Y. The use of alcohol use disorders identification test (AUDIT) in detecting alcohol use disorder and risk drinking in the general population: validation of AUDIT using schedules for clinical assessment in neuropsychiatry. *Alcohol Clin. Exp. Res.* **39**, 158–165 (2015).
56. Hildebrand, M. The psychometric properties of the Drug Use Disorders Identification Test (DUDIT): a review of recent research. *J. Subst. Abuse. Treat.* **53**, 52–59 (2015).
57. Currie, S. R., Hodgins, D. C. & Casey, D. M. Validity of the Problem Gambling Severity Index interpretive categories. *J. Gambl. Stud.* **29**, 311–327 (2013).
58. Bollen, K. A. et al. *Testing Structural Equation Models* (Sage, 1993).
59. Hancock, G. R. & Mueller, R. O. in *Structural Equation Modeling: Present and Future: A Festschrift in Honor of Karl Jöreskog* (eds Cudeck R., Du Toit S. & Sörbom D.) 195–216 (Scientific Software International, 2001).

60. Fornell, C. & Larcker, D. F. Evaluating structural equation models with unobservable variables and measurement error. *J. Mark. Res.* **18**, 39–50 (1981).
61. Meredith, W. Measurement invariance, factor-analysis and factorial invariance. *Psychometrika* **58**, 525–543 (1993).
62. Vandenberg, R. J. Toward a further understanding of and improvement in measurement invariance methods and procedures. *Organ. Res. Methods* **5**, 139–158 (2002).
63. Muthén, L. K. & Muthén, B. O. *Mplus User's Guide: The Comprehensive Modeling Program for Applied Researchers* 1st edn (Muthén & Muthén, 1998).
64. Silvia, E. S. M. & MacCallum, R. C. Some factors affecting the success of specification searches in covariance structure modeling. *Multivar. Behav. Res.* **23**, 297–326 (1988).
65. Benjamini, Y. & Hochberg, Y. Controlling the false discovery rate – a practical and powerful approach to multiple testing. *J. R. Stat. Soc. Ser. B* **57**, 289–300 (1995).
66. Marsh, H. W., Hau, K. T. & Wen, Z. L. In search of golden rules: comment on hypothesis-testing approaches to setting cutoff values for fit indexes and dangers in overgeneralizing Hu and Bentler's (1999) findings. *Struct. Equ. Modeling* **11**, 320–341 (2004).
67. Hayduk, L., Cummings, G., Boadu, K., Pazderka-Robinson, H. & Boulianne, S. Testing! testing! one, two, three – Testing the theory in structural equation models! *Pers. Individ. Differ.* **42**, 841–850 (2007).
68. Byrne, B. M., Shavelson, R. J. & Muthén, B. Testing for the equivalence of factor covariance and mean Structures – the issue of partial measurement invariance. *Psychol. Bull.* **105**, 456–466 (1989).
69. Kline, R. B. *Principles and Practice of Structural Equation modeling* 4th edn (Guilford Press, 2015).
70. Wagenmakers, E. J. A practical solution to the pervasive problems of *p* values. *Psychon. Bull. Rev.* **14**, 779–804 (2007).
71. Bollen, K. A. Modeling strategies: in search of the holy grail. *Struct. Equ. Modeling* **7**, 74–81 (2000).
72. Putnick, D. L. & Bornstein, M. H. Measurement invariance conventions and reporting: the state of the art and future directions for psychological research. *Dev. Rev.* **41**, 71–90 (2016).
73. Yoon, M. & Kim, E. S. A comparison of sequential and nonsequential specification searches in testing factorial invariance. *Behav. Res. Methods* **46**, 1199–1206 (2014).
74. Hair, J. F., Black, W. C., Babin, B. J. & Anderson, R. E. *Multivariate Data Analysis* 7th edn (Pearson, 2014).
75. Wasserstein, R. L., Schirm, A. L. & Lazar, N. A. Moving to a world beyond ' $p < 0.05$ '. *Am. Stat.* **73**, 1–19 (2019).
76. Joreskog, K. G., Olsson, U. H. & Wallentin, F. Y. *Multivariate Analysis with LISREL* 1st edn (Springer, 2016).
77. McArdle, J. J. Causal-modeling applied to psychonomic systems simulation. *Behav. Res. Methods Instrum.* **12**, 193–209 (1980).
78. Shrout, P. E. & Fleiss, J. L. Intraclass correlations: uses in assessing rater reliability. *Psychol. Bull.* **86**, 420–428 (1979).
79. Cicchetti, D. V. & Sparrow, S. A. Developing criteria for establishing interrater reliability of specific items – applications to assessment of adaptive-behavior. *Am. J. Ment. Defic.* **86**, 127–137 (1981).

Acknowledgements

This study was funded by the Australian Research Council through grants LP150100770 and DP180100145 (chief investigators: A.V.-G., M.A.B., and D.I.L.). A.V.-G. was funded by an Australian Medical Research Future Fund Career Development Fellowship (level 2, MRF1141214). J.T. was supported by National Health and Medical Research Council (NHMRC) project grants 1002458 and 1046054. M.A.B. was funded by an NHMRC Senior Research Fellowship (level B). Torus Games designed the CIS as a contractor, but they have also provided in-kind support and ideas, and the authors thank K. MacIntosh (Torus Games Head of Production) for support.

Author contributions

A.V.-G., M.A.B. and D.I.L. conceptualized the study designs. N.K., N.M., A.A. and J.K. collected the data. J.T., N.M., N.K. and K.V. analysed the data. J.T., N.K. and N.M. prepared the figures and tables. A.V.-G., J.T., N.K. and N.M. wrote the initial manuscript draft. All authors contributed to writing and revising the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at <https://doi.org/10.1038/s41562-021-01127-3>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41562-021-01127-3>.

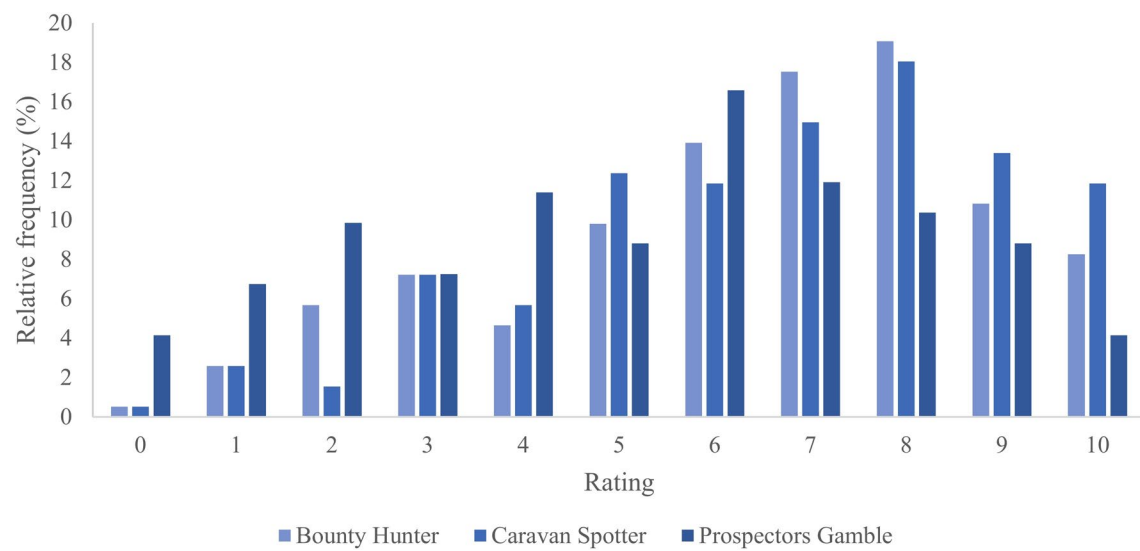
Correspondence and requests for materials should be addressed to A.V.-G.

Peer review information *Nature Human Behaviour* thanks Robert Leeman and A. Zeynep Enkavi for their contributions to the peer review of this work.

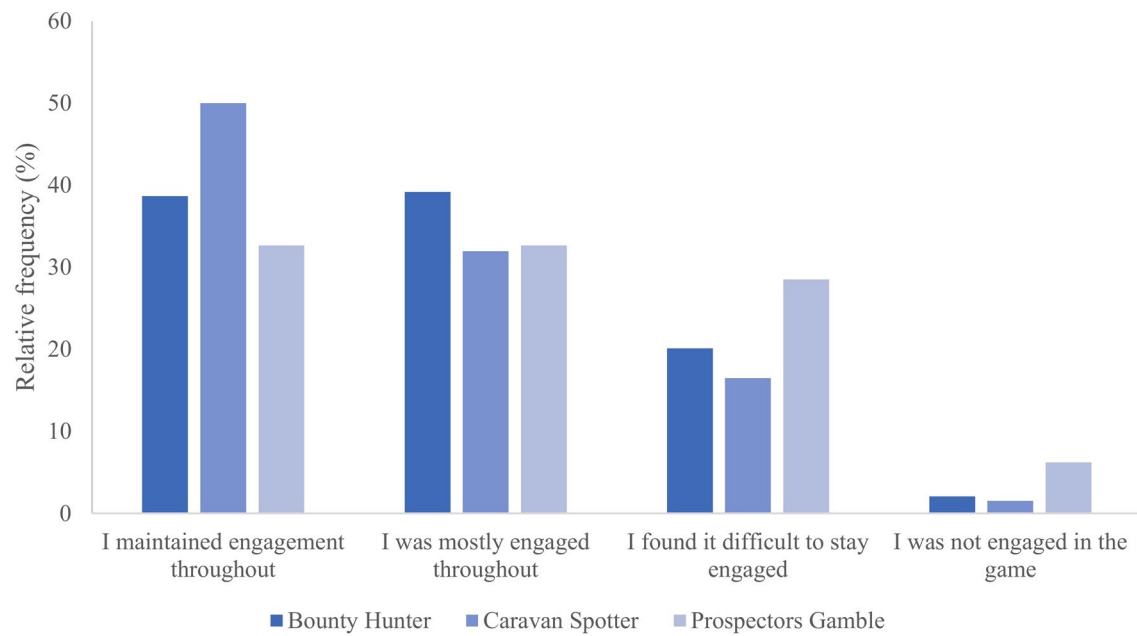
Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

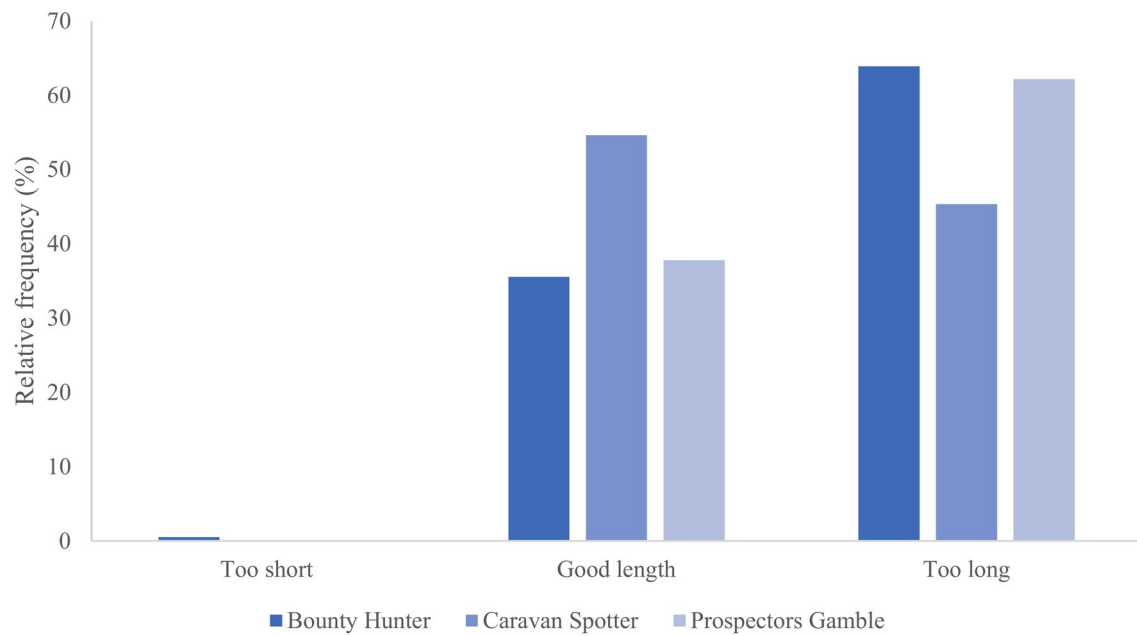
© The Author(s), under exclusive licence to Springer Nature Limited 2021



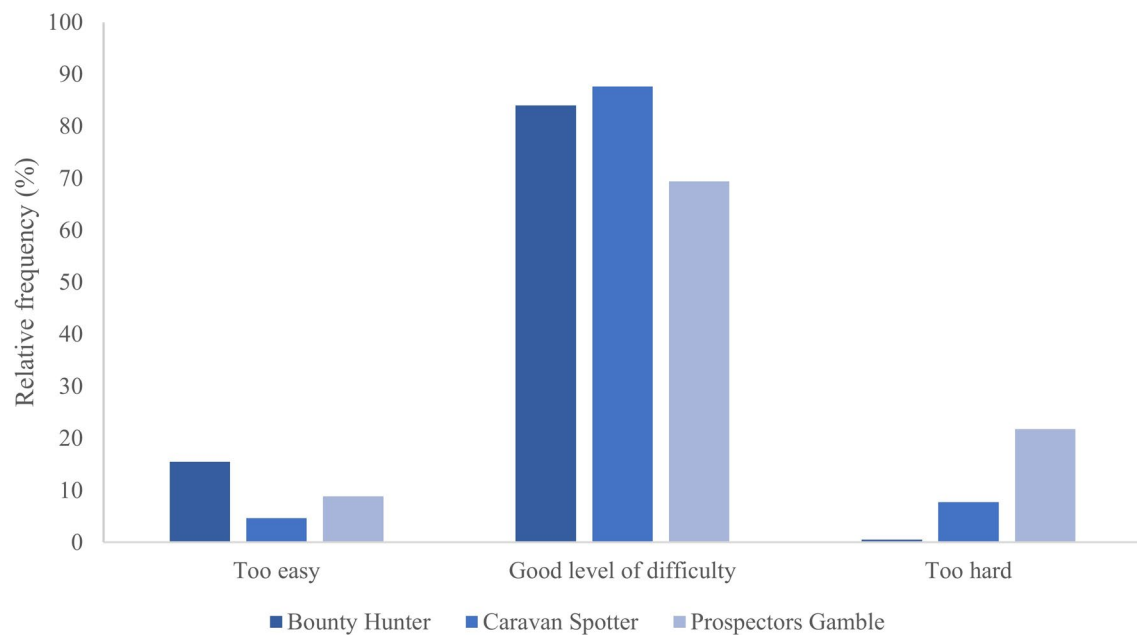
Extended Data Fig. 1 | Relative frequencies of enjoyment ratings for the three CIS tasks.



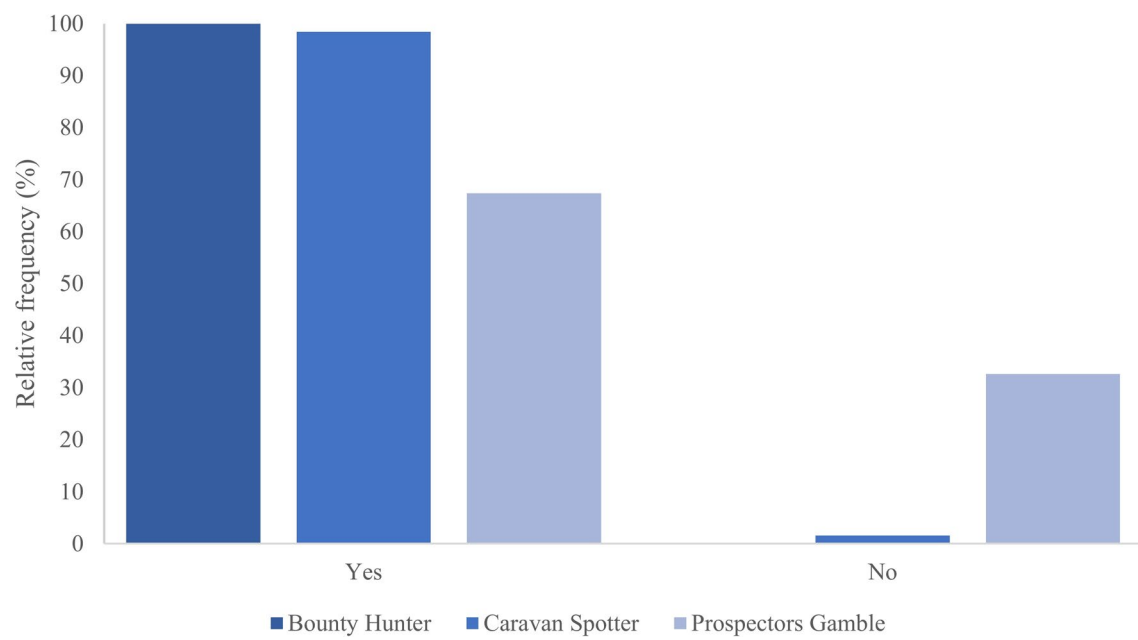
Extended Data Fig. 2 | Relative frequencies of engagement ratings for the three CIS tasks.



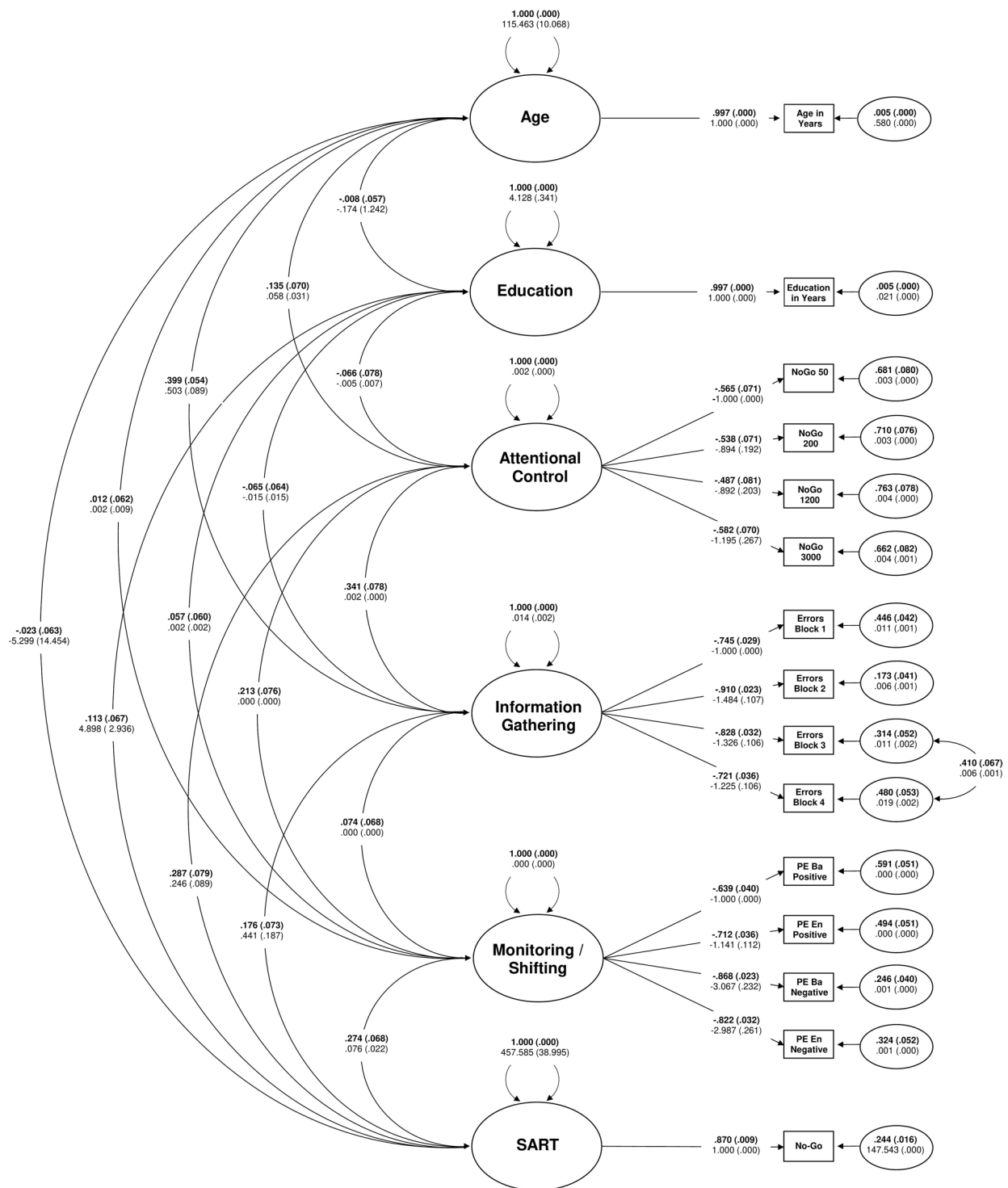
Extended Data Fig. 3 | Relative frequencies of duration ratings for the three CIS tasks.



Extended Data Fig. 4 | Relative frequencies of difficulty ratings for the three CIS tasks.

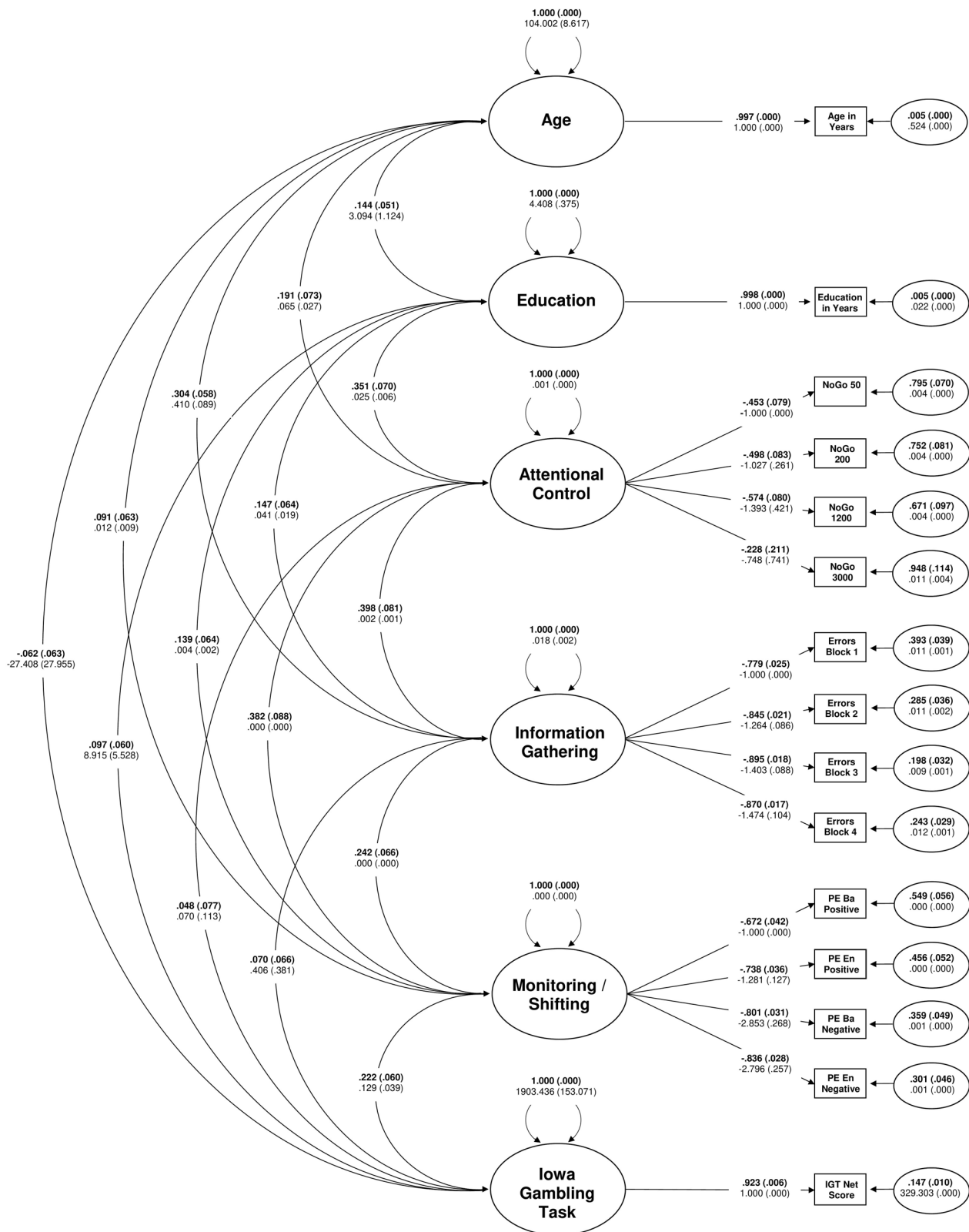


Extended Data Fig. 5 | Relative frequencies of clarity ratings for instructions for the three CIS tasks.



Extended Data Fig. 6 | See next page for caption.

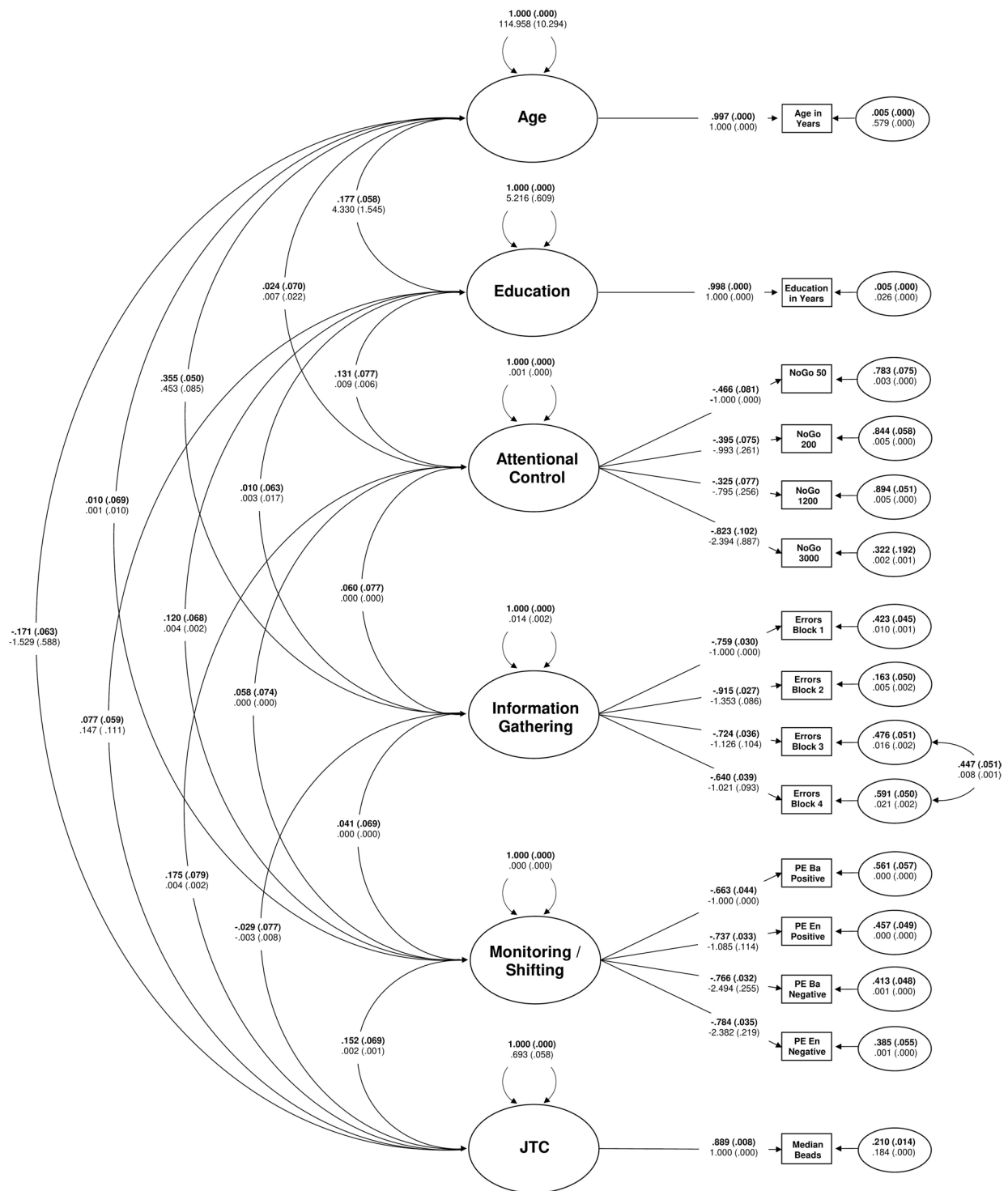
Extended Data Fig. 6 | Confirmatory factor analysis model of latent correlation between the three cognitive constructs measured by the CIS and performance on the SART. Model fit statistics were $\chi^2(77) = 101.010$, $p = .129$; RMSEA = .032 [90%CI = .009, .049]; CFI = .982; SRMR = .038. *Note.* NoGo50 = Commission errors on no-go trials on Bounty Hunter with 50 ms stimulus onset asynchrony; NoGo200 = Commission errors on no-go trials on Bounty Hunter with 200 ms stimulus onset asynchrony; NoGo1200 = Commission errors on no-go trials on Bounty Hunter with 1200 ms stimulus onset asynchrony; NoGo3000 = Commission errors on no-go trials on Bounty Hunter with 3000 ms stimulus onset asynchrony; Errors Block 1 = Fast identification errors on Caravan Spotter Block 1; Errors Block 2 = Fast identification errors on Caravan Spotter Block 2; Errors Block 3 = Fast identification errors on Caravan Spotter Block 3; Errors Block 4 = Fast identification errors on Caravan Spotter Block 4. PEBa Positive = Perseverative errors on Prospector's Gamble basic positive feedback on previous trial; PEEn Positive = Perseverative errors on Prospector's Gamble enhanced positive feedback on previous trial; PEBa Negative = Perseverative errors on Prospector's Gamble basic negative feedback on previous trial; PEEn Negative = Perseverative errors on Prospector's Gamble enhanced negative feedback on previous trial. SART = sustained attention to response task. No-Go = successful inhibitions on no-go trials of the SART. Factor scaling was performed using the reference variable method. Standardised estimates appear in bold typeface. Unstandardised estimates are in normal typeface below. Bootstrapped standard errors (10,000 posterior draws) appear in brackets. Threshold for statistical significance for associations between SART No-Go Commission Errors and the CIS factors using the Adjusted Bonferroni procedure ($AB2; \alpha_{per\ test} = \frac{\alpha_{familywise}}{1 - \sqrt{|I_j|}}$) was $\alpha < .025$.



Extended Data Fig. 7 | See next page for caption.

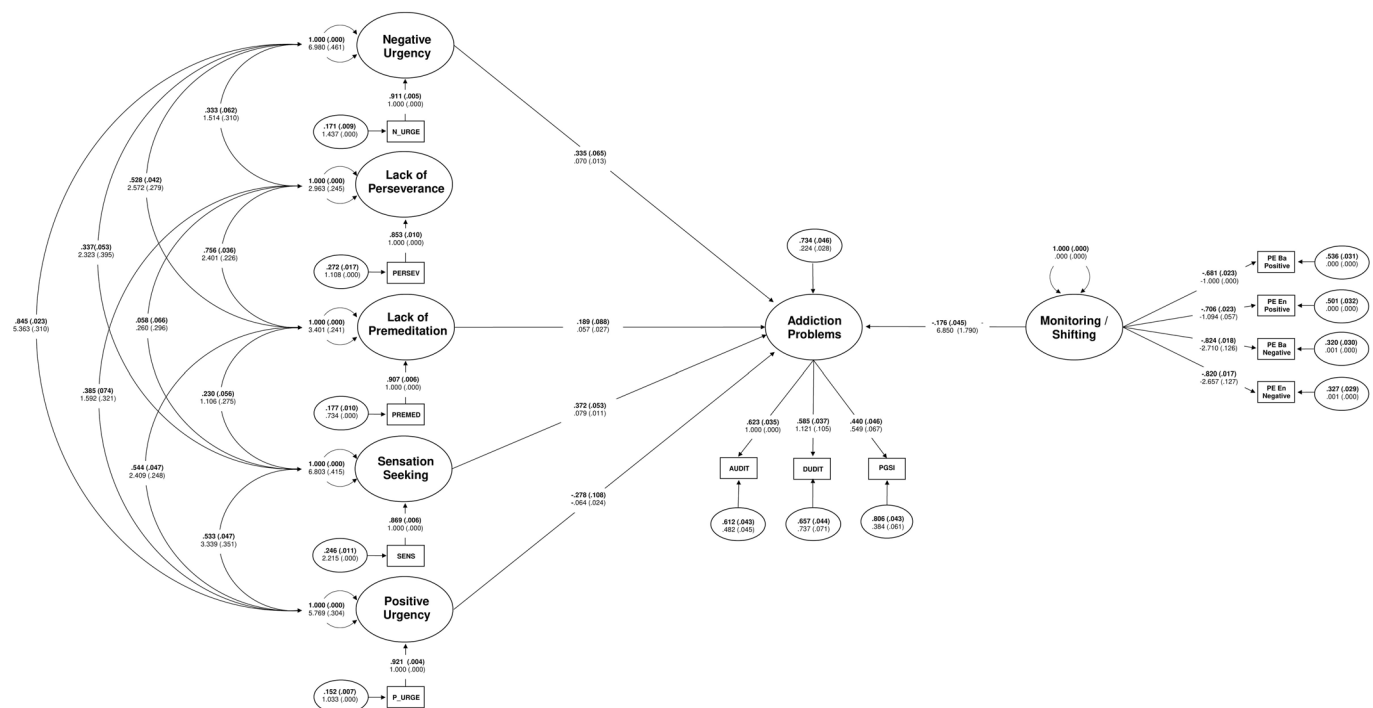
Extended Data Fig. 7 | Confirmatory factor analysis model of latent correlation between the three cognitive constructs measured by the CIS and performance on the IGT. Model fit statistics were $\chi^2(78) = 95.073$, $p = .351$; RMSEA = .027 [90%CI = .000, .044]; CFI = .988; SRMR = .034. *Note.*

NoGo50 = Commission errors on no-go trials on Bounty Hunter with 50 ms stimulus onset asynchrony; NoGo200 = Commission errors on no-go trials on Bounty Hunter with 200 ms stimulus onset asynchrony; NoGo1200 = Commission errors on no-go trials on Bounty Hunter with 1200 ms stimulus onset asynchrony; NoGo3000 = Commission errors on no-go trials on Bounty Hunter with 3000 ms stimulus onset asynchrony; Errors Block 1 = Fast identification errors on Caravan Spotter Block 1; Errors Block 2 = Fast identification errors on Caravan Spotter Block 2; Errors Block 3 = Fast identification errors on Caravan Spotter Block 3; Errors Block 4 = Fast identification errors on Caravan Spotter Block 4. PEBa Positive = Perseverative errors on Prospector's Gamble basic positive feedback on previous trial; PEEn Positive = Perseverative errors on Prospector's Gamble enhanced positive feedback on previous trial; PEBa Negative = Perseverative errors on Prospector's Gamble basic negative feedback on previous trial; PEEn Negative = Perseverative errors on Prospector's Gamble enhanced negative feedback on previous trial. IGT = Iowa gambling task. IGT net score = number of 'safe deck' choices subtracted from 'risky deck' choices. Factor scaling was performed using the reference variable method. Standardised estimates appear in bold typeface. Unstandardised estimates are in normal typeface below. Bootstrapped standard errors (10,000 posterior draws) appear in brackets. Threshold for statistical significance for associations between IGT Net Score and the other latent variables using the Adjusted Bonferroni procedure (AB2; $\alpha_{per\ test} = \frac{\alpha_{familywise}}{1 - \sqrt{k}}$) was $\alpha < .027$.

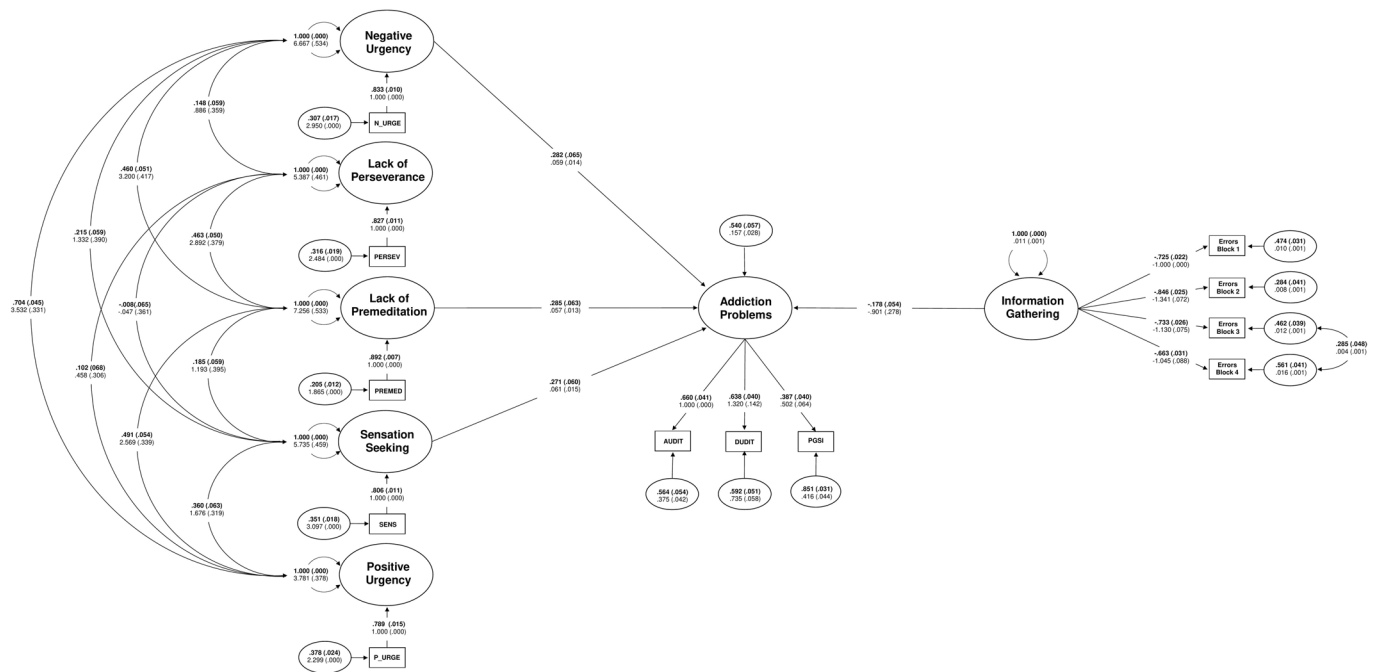


Extended Data Fig. 8 | See next page for caption.

Extended Data Fig. 8 | Confirmatory factor analysis model of latent correlation between the three cognitive constructs measured by the CIS and performance on the JTC task. Model fit statistics were $\chi^2(77) = 102.834$, $p = .069$; RMSEA = .033 [90%CI = .012, .049]; CFI = .978; SRMR = .044. *Note.* NoGo50 = Commission errors on no-go trials on Bounty Hunter with 50 ms stimulus onset asynchrony; NoGo200 = Commission errors on no-go trials on Bounty Hunter with 200 ms stimulus onset asynchrony; NoGo1200 = Commission errors on no-go trials on Bounty Hunter with 1200 ms stimulus onset asynchrony; NoGo3000 = Commission errors on no-go trials on Bounty Hunter with 3000 ms stimulus onset asynchrony; Errors Block 1 = Fast identification errors on Caravan Spotter Block 1; Errors Block 2 = Fast identification errors on Caravan Spotter Block 2; Errors Block 3 = Fast identification errors on Caravan Spotter Block 3; Errors Block 4 = Fast identification errors on Caravan Spotter Block 4. PEBa Positive = Perseverative errors on Prospector's Gamble basic positive feedback on previous trial; PEEb Positive = Perseverative errors on Prospector's Gamble enhanced positive feedback on previous trial; PEBa Negative = Perseverative errors on Prospector's Gamble basic negative feedback on previous trial; PEEb Negative = Perseverative errors on Prospector's Gamble enhanced negative feedback on previous trial. JTC = Jumping to conclusions task. Median Beads = median number of beads drawn before a decision is made (aggregated across blocks with probabilities 85:15 and 60:40). Factor scaling was performed using the reference variable method. Standardised estimates appear in bold typeface. Unstandardised estimates are in normal typeface below. Bootstrapped standard errors (10,000 posterior draws) appear in brackets. Threshold for statistical significance for associations with the median beads drawn on the JTC tasks and the other latent variables using the Adjusted Bonferroni procedure (AB2; $\alpha_{per\ test} = \frac{\alpha_{familywise}}{k^{1-\sqrt{|\hat{\gamma}|}}}$ was $\alpha < .024$.



Extended Data Fig. 9 | Incremental criterion validity model with Addiction Problems factor regressed onto four of the five UPPS-P impulsivity factors and the CIS Monitoring / Shifting factor in the AMT calibration subsample. Model fit statistics were $\chi^2(107) = 125.537$, $p = .330$; RMSEA = .018 [90%CI = .000, .030]; CFI = .995; SRMR = .033. $n = 510$. $R^2 = .266$, $SE = .052$, one-tailed $p < .001$; $\Delta R^2 = .013$. Note. N_URGE = Negative Urgency raw scores from the UPPS-P; P_URGE = Positive Urgency raw scores from the UPPS-P; PREM = Lack of Premeditation raw scores from the UPPS-P; SENS = Sensation Seeking raw scores from the UPPS-P; PERSEV = Lack of Perseverance raw scores from the UPPS-P; PE Ba Positive = Perseverative errors on Prospector's Gamble basic positive feedback on previous trial; PE En Positive = Perseverative errors on Prospector's Gamble enhanced positive feedback on previous trial; PE Ba Negative = Perseverative errors on Prospector's Gamble basic negative feedback on previous trial; PE En Negative = Perseverative errors on Prospector's Gamble enhanced negative feedback on previous trial. AUDIT = Alcohol Use Disorder Identification Test raw scores logarithmic transformed; DUDIT = Drug Use Disorder Identification Test raw scores logarithmic transformed; PGSI = Problem Gambling Severity Index raw scores logarithmic transformed. Factor scaling was performed using the reference variable method. Standardised estimates appear in bold typeface. Unstandardised estimates are in normal typeface below. Bootstrapped standard errors (10,000 posterior draws) appear in brackets. The probability value of the standardised beta coefficient for the regression of Addiction Problems on Lack of Perseverance was $p = .111$. The latent correlations between the Monitoring / Shifting factor and the five self-reported impulsivity factors have been omitted from the figure for clarity. These were Negative Urgency ($\phi = -.134$, $SE = .038$, [95%BCI = -.201, -.050], $p < .001$); Lack of Perseverance ($\phi = -.037$, $SE = .041$, [95%BCI = -.128, .036], $p = .377$); Lack of Premeditation ($\phi = -.154$, $SE = .039$, [95%BCI = -.228, -.071], $p < .001$); Sensation Seeking ($\phi = -.110$, $SE = .040$, [95%BCI = -.185, -.028], $p = .006$); Positive Urgency ($\phi = -.278$, $SE = .037$, [95%BCI = -.348, -.204], $p < .001$).



Extended Data Fig. 10 | Incremental criterion validity model with Addiction Problems factor regressed onto Negative Urgency, Lack of Premeditation, and Sensation Seeking and Information Gathering factor in the community and clinical sample. Model fit statistics were $\chi^2(39) = 57.482$, $p = .082$; RMSEA = .029 [90%CI = .010, .044]; CFI = .988; SRMR = .028. $n = 578$. $R^2 = .460$, $SE = .057$, one-tailed $p < .001$; $\Delta R^2 = .028$. The latent correlations between the Information Gathering factor and the five self-reported impulsivity factors have been omitted from the figure for clarity. These were Negative Urgency ($\phi = -.172$, $SE = .054$, [95%BCI = $-.272$, $-.059$], $p = .001$); Lack of Perseverance ($\phi = -.080$, $SE = .056$, [95%BCI = $-.192$, $-.029$], $p = .155$); Lack of Premeditation ($\phi = -.195$, $SE = .049$, [95%BCI = $-.284$, $-.093$], $p < .001$); Sensation Seeking ($\phi = -.215$, $SE = .057$, [95%BCI = $-.320$, $-.099$], $p < .001$); and Positive Urgency ($\phi = -.205$, $SE = .058$, [95%BCI = $-.315$, $-.087$], $p < .001$). Note. N_URGE = Negative Urgency raw scores from the UPPS-P; P_URGE = Positive Urgency raw scores from the UPPS-P; PREM = Lack of Premeditation raw scores from the UPPS-P; SENS = Sensation Seeking raw scores from the UPPS-P; PERSEV = Lack of Perseverance raw scores from the UPPS-P; PEBa Positive = Perseverative errors on Prospector's Gamble basic positive feedback on previous trial; PEE Positive = Perseverative errors on Prospector's Gamble enhanced positive feedback on previous trial; PEBa Negative = Perseverative errors on Prospector's Gamble basic negative feedback on previous trial; PEE Negative = Perseverative errors on Prospector's Gamble enhanced negative feedback on previous trial. AUDIT = Alcohol Use Disorder Identification Test raw scores logarithmic transformed; DUDIT = Drug Use Disorder Identification Test raw scores logarithmic transformed; PGSI = Problem Gambling Severity Index raw scores logarithmic transformed. Factor scaling was performed using the reference variable method. Standardised estimates appear in bold typeface. Unstandardised estimates are in normal typeface below. Bootstrapped standard errors (10,000 posterior draws) appear in brackets.

Reporting Summary

Nature Research wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Research policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐ ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☐ ☒ A description of all covariates tested
- ☐ ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☐ ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☐ ☒ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☐ ☒ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection The Amazon MTurk platform was used to recruit participants. Qualtrics software was used to collect survey data. Inquisit software was used to collect data from the Sustained Attention to Response Task (SART) the Jumping to Conclusions (JTC) task and the Iowa Gambling Task (IGT). Custom-made code was used to administer the Cognitive Impulsivity Suite (CIS).

Data analysis Mplus 8.3 code was used to estimate the structural equation models presented in this manuscript

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

Data obtained from cognitive testing and self-report and analysed for this manuscript can be accessed via the Open Science Framework [<https://osf.io/7qrv5/>].

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☒ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	Cross sectional design examining the factor structure of a novel cognitive measure of impulsivity (CIS); its convergent validity against a gold standard survey measure of impulsivity (the Short Urgency, Premeditation, Perseverance, Sensation Seeking - Positive urgency scale [SUPPS-P]); and its criterion related validity against self-reports of real-world impulsive behaviours including alcohol use (Alcohol Use Identification Test [AUDIT]), drug use (Drug Use Identification Test [DUDIT]), and gambling (Problem Gambling Screening Inventory [PGSI]).
Research sample	Three independent samples: an online community sample (N = 1,056), a face-to-face community sample (n = 63), and a replication online community and clinical sample (N = 578)
Sampling strategy	We used crowdsourcing to recruit the initial online community sample (via Amazon MTurk) and community advertisement (flyers, websites such as Gumtree) as well as an Institution hosted online system (SONA) to recruit participants for the face-to-face community sample. The replication sample was recruited via two different routes: community-dwelling participants were recruited through an participant database for the Monash Brain and Behaviour Project and through online advertisements via the project Facebook page, and clinical participants were recruited through Counselling Online services, an online treatment platform. Desired sample size was determined to be 250 – 300. Sample size for structural equation modelling was based on two complementary approaches: 1) the minimum participant-to-parameter (N:q) ratio (Jackson, 2003; Kline, 2015); and 2) power for testing the alternative hypothesis of ‘not close fit’ ($\epsilon > .05$) against the null hypothesis of ‘close fit’ ($\epsilon = .00$) of the model reproduced covariance matrix to the population covariance matrix based on the Root Mean Square Error of Approximation (RMSEA) and 90% confidence interval (MacCallum, Browne, & Sugawara, 1996). The N:q ratio determines the minimum sample required to parameterise the most complex models (i.e. models with the most freely estimated parameters). The recommended minimum N:q ratio is between 5:1 and 10:1, as models with fewer than 5 cases per parameter produce less trustworthy estimates, standard errors, and confidence intervals (Kline, 2015). The models planned for replication consist of approximately 27 - 50 freely estimated parameters, such that a sample size of 250 - 300 would provide participant-to-parameter ratios of between 5:1 and 11:1, which is considered adequate for model testing. Statistical power for the RMSEA is determined by model degrees of freedom (df) (i.e. the number of unique variance-covariance terms in the model minus the number of freely estimated parameters) (MacCallum et al., 1996). The minimum sample size required for sufficient power ($\alpha < .05$; $1 - \beta = 0.80$) to adjudicate between the null and alternative hypothesis based on the RMSEA for models with the fewest df (i.e. 51) was N = 232 (Preacher & Coffman, May 2006). References: Jackson, D. L. (2003). Revisiting sample size and number of parameter estimates: Some support for the N : q hypothesis. <i>Structural equation modeling</i>, 10, 128 - 141. doi:10.1207/s15328007sem1001_6 Kline, R. B. (2015). <i>Principles and practice of structural equation modeling</i> (4th ed.). New York: The Guilford Press. MacCallum, R. C., Browne, M. W., & Sugawara, H. M. (1996). Power analysis and determination of sample size for covariance structure modeling. <i>Psychological methods</i>, 1, 130-149. Preacher, K. J., & Coffman, D. L. (May 2006). Computing power and minimum sample size for RMSEA [Computer software]. . Available from http://quantpsy.org/.
Data collection	The testing of the original online community sample was facilitated by TurkPrime, an online crowdsourcing platform that integrates with the Amazon MTurk platform, which we recruited the participants from. Participants were first directed to a survey hosted by Qualtrics where they were provided with a digital Explanatory Statement and gave informed consent. Second, they were redirected from Qualtrics to complete the CIS, then redirected back to Qualtrics to complete a demographics survey and self-report measures of alcohol use, drug use, and gambling (AUDIT, DUDIT, PGSI). Participants completed a demo version of the CIS before starting the tasks to minimise technical errors and performance anxiety. Finally, participants were directed to Inquisit, a study platform hosted by Millisecond (www.millisecond.com), where they first completed a self-report measure of trait impulsivity (short UPPS-P) before being randomly allocated to complete one of three cognitive tasks (i.e., JTC, IGT or SART). The face-to-face community sample data was collected using research-dedicated computers in the Monash School of Psychological Sciences. Only participants and the researcher / assessor were present. The online replication community and clinical sample followed the same procedures as the original MTurk sample, although we used personalised web-links instead of the MTurk platform. No formal blinding procedure was conducted.

Timing

For the online community (MTurk) sample: The data collection period was 27/09/2018 - 11/10/2018.

For the face-to-face community sample: The data collection period was 12/07/2018 - 08/03/2019.

For the replication community and clinical sample: The data collection period was 15/07/2020 - 25/10/2020.

Data exclusions

We excluded participants who had suboptimal performance in the CIS, as determined by a set of a priori criteria based on those published by Eisenberg et al. (2019) Nat Commun. 2019;10(1):2319. doi: 10.1038/s41467-019-10301-1. The number of participants excluded in each sample and for the different tasks was as follows:

Online community sample (MTurk):

942 completes

Excluded:

Bounty Hunter: n = 48

Caravan Spotter: n = 71

Prospector's Gamble: n = 35

Face-to-face community sample:

60 completes for both sessions

Excluded:

Session 1

Bounty Hunter: 1

Caravan Spotter: 0

Prospector's Gamble: 0

Session 2

Bounty Hunter: 1

Caravan Spotter: 1

Prospector's Gamble: 0

Replication sample / Community subsample:

312 completes

Excluded:

Bounty Hunter: 10

Caravan Spotter: 6

Prospector's Gamble: 3

Replication sample / Clinical subsample:

259 completes

Bounty Hunter: 22

Caravan Spotter: 11

Prospector's Gamble: 11

Non-participation

Response rates following consent for the different samples were as follows:

Online community sample (MTurk): 88.2%

Face-to-face community sample: 95.2%

Replication sample / Community subsample: 85.1%

Replication sample / Clinical subsample: 46%

Randomization

Non-applicable. The study was cross-sectional / correlational, there was no allocation of participants to groups or planned group comparisons

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Human research participants

Policy information about [studies involving human research participants](#)

Population characteristics	See above
Recruitment	Participants in the initial online community sample were recruited through MTurk, and community-dwelling participants in the replication sample were recruited through a participants' database of an existing project, so in both cases there is a risk of self-selection bias as these participants had prior research experience. Participants in the face-to-face sample were recruited through community advertisement and participants in the clinical subsample of the replication sample via an online treatment portal.
Ethics oversight	The Monash University Human Research Ethics Committee approved the study (MUHREC project numbers 7536 and 11999) and all participants signed an informed consent form

Note that full information on the approval of the study protocol must also be provided in the manuscript.